

ADVERSARIAL ATTACKS LEVERAGE INTERFERENCE BETWEEN FEATURES IN SUPERPOSITION

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ABSTRACT

Fundamental questions remain about when and why adversarial examples (AExs) arise across the broad range of neural networks (NNs) in use today, with competing views characterising them either as artifacts of the irregularities in the decision landscape or as products of sensitivity to non-robust input features. In this paper, we instead argue that adversarial vulnerability can be understood, quite generally, as a consequence of *efficient* information encoding in NNs. Specifically, we show how superposition – a widespread strategy in which networks represent more features than they have dimensions – gives rise to arrangements of latent representations that adversaries across attack settings can exploit. We demonstrate that adversarial perturbations leverage interference between superposed features, so that attack patterns become predictable from the underlying feature arrangements. Our framework offers a unifying mechanistic explanation for two widely observed phenomena: adversarial attack transferability between models with related training regimes and class-specific vulnerability patterns. In synthetic settings with precisely controlled superposition, we establish that superposition *suffices* to create adversarial vulnerability, isolating the mechanism in a form that carries over to richer models. We then show that the same principles persist in a vision transformer (ViT) trained on CIFAR-10, and, more broadly, across additional architectures and representational regimes. Taken together, these findings indicate that adversarial vulnerability can be a natural byproduct of the representational compression that pervades modern networks, rather than a flaw in the learning process or a fixed property of non-robust inputs. We release our implementation through a software repository.

1 INTRODUCTION

Despite extensive research on AExs across tasks, architectures, and data modalities (Szegedy et al., 2014; Goodfellow et al., 2014), there remains no consensus on their fundamental cause. We still lack a general explanation for how minimal perturbations so drastically alter model predictions, leaving us unable to anticipate which perturbations succeed, to explain why attacks transfer so readily between models, or to design defences that address the underlying cause rather than a single attack.

Existing explanations broadly fit into two camps (Nakkiran, 2019). The *bug* perspective attributes vulnerability to factors including learning flaws (Schmidt et al., 2018; Nakkiran, 2019) or geometric properties of high-dimensional decision boundaries (Fawzi et al., 2016). Work in this domain typically provides statistical bounds on properties such as capacity (Bubeck & Sellke, 2021) or Lipschitz constraints (Hein & Andriushchenko, 2017), but such bounds do not identify which specific perturbations succeed. The *feature* perspective argues AExs exploit predictive but non-robust statistical patterns in data (Ilyas et al., 2019), reframing the debate around how the underlying data relates to human perception. It treats these non-robust features as fixed properties of the data rather than as products of the encoding strategies a network adopts. Neither view reconciles how representational constraints interact with data semantics, a gap that spans the many settings in which AExs appear.

In this paper, we bridge this divide through a mechanistic interpretability perspective, showing how adversarial vulnerability emerges quite generally from the interaction between architectural constraints and data semantics (*i.e.* human-interpretable properties of the data). We show that AExs can arise from interference between learned representations in NNs. Adversarial attacks systematically exploit superposed latent structure – a mechanism that normally enables additional representational capacity – to craft effective perturbations that manipulate model outputs. Our analysis reveals a

mechanistic pathway that applies wherever superposition is present: input correlations constrain feature arrangements, these arrangements determine interference patterns, and these interference patterns in turn dictate attack characteristics and transferability. This framework lets us predict which perturbations succeed and why they transfer between models.

Our account of AExs draws on recent theories about how networks encode and process information, namely the linear representation hypothesis (LRH) (Park et al., 2024) and the theory of superposition (Elhage et al., 2022). The LRH posits that *input features*—fundamental abstractions of data—are represented as linear directions in a network’s representation space. It is hypothesised that NNs can represent significantly more of these features than they have neurons through superposition, enabling networks to efficiently pack many features into shared dimensions at the cost of introducing interference. Such interference means perturbing one feature can affect others in non-obvious ways. Because superposition is increasingly understood to be pervasive across model families and scales, this paper investigates whether the resulting interference creates vulnerabilities that AExs exploit, and what broadly applicable insights this offers for understanding adversarial phenomena. Our primary contributions are:

- We present a general framework reconciling architectural constraints with feature properties in adversarial vulnerability. Using simple, controllable models, we demonstrate how data properties induce specific superposition geometries that directly determine the realised adversarial perturbations, in a way that isolates a mechanism expected to recur in larger systems. We show that interference patterns offer insight into attack transferability and class-specific susceptibility.
- We demonstrate that these findings persist in a ViT trained on CIFAR-10 with an engineered bottleneck layer to induce superposition, and that the same trends recur across additional architectures.
- We show superposition to be sufficient, but not necessary, for adversarial vulnerability. In identifying this, we establish algorithmic brittleness as a distinct vulnerability mechanism, broadening the representation-level view beyond superposition alone.
- We demonstrate that a mechanistic understanding of learned representations enables informed adversarial attack construction, underpinning our broader argument for semantically-informed defences.

2 BACKGROUND

We now provide an overview of the tools we use in our analysis: the LRH and superposition. Formally, let $\mathbf{x} \in \mathcal{X}$ denote the input to a NN and $\mathbf{h}^{(l)} \in \mathbb{R}^{d_l}$ the activation vector of the l -th layer.

Linear representation hypothesis (LRH). The LRH posits that NNs, across architectures and modalities, represent many variables of their computation—such as semantic properties of their inputs—as linear directions in their activation space, which can serve as abstractions for reasoning (Park et al., 2024; Guerner et al., 2023). This bias towards linear features is hypothesised to arise because linear separability allows networks to easily recognise and manipulate features, and because dot products with subsequent layer weights efficiently process such directional features. A growing body of evidence, spanning language and vision models, supports this view (Gurnee et al., 2023; Park et al., 2024). Let $\mathcal{C} = \{c_1, \dots, c_M\}$ denote a set of M semantically meaningful latent features (e.g. concepts like “presence of shape,” or “indoor vs. outdoor”). Formally:

Definition 1 (Linear Representation Hypothesis (LRH)) *A neural network layer with activations $\mathbf{h}^{(l)} \in \mathbb{R}^{d_l}$ satisfies the LRH if it represents the latent features $\mathcal{C} = \{c_1, \dots, c_M\}$ as linear directions $\{\mathbf{v}_j\}_{j=1}^M \subset \mathbb{R}^{d_l}$ such that:*

$$\mathbf{h}^{(l)}(\mathbf{x}) \approx \sum_{j=1}^M a_j(\mathbf{x}) \mathbf{v}_j$$

where $a_j(\mathbf{x}) \in \mathbb{R}$ represents the activation magnitude of feature c_j , and $\mathbf{v}_j$ is the corresponding linear direction. The features are **linearly accessible** (Costa et al., 2025): inputs $\mathbf{x}_0, \mathbf{x}_1 \in \mathcal{X}$ that differ mainly in the value of feature c_j while holding other features approximately constant (i.e., $a_j(\mathbf{x}_1) - a_j(\mathbf{x}_0) = k$ and $|a_i(\mathbf{x}_1) - a_i(\mathbf{x}_0)| < \lambda$ for all $i \neq j$ and some small $\lambda > 0$) satisfy:

$$\mathbf{h}^{(l)}(\mathbf{x}_1) - \mathbf{h}^{(l)}(\mathbf{x}_0) \approx k \mathbf{v}_j$$

where k reflects the change in c_j .

Superposition and sparsity. NNs are, in general, capable of representing many more latent features than there are available dimensions in activation space: $M > d_l$. For example, LLMs can reference far more place names than they have residual stream dimensions, and analogous over-representation is observed across model families. One framework for analysing this broad phenomenon is superposition, according to which networks use an overcomplete and non-orthogonal set of feature directions $\{\mathbf{v}_j\}_{j=1}^M$. This leverages the fact that $2^{\Theta(d\epsilon^2)}$ almost orthogonal vectors ($< \epsilon$ cosine similarity) can be represented in d -dimensional space (Tkocz, 2012), and that sparse vectors can be accurately recovered after projection into lower-dimensional spaces (Elhage et al., 2022; Bereska & Gavves, 2024; Sawmya et al., 2025). This creates two challenges that accompany superposition wherever it appears. First, *polysemanticity* (Scherlis et al., 2022; Lecomte et al., 2023) emerges, where individual neurons contribute to multiple distinct features, so that a neuron’s activation no longer corresponds to a single interpretable concept. Second, non-orthogonal feature directions create *interference* between features – activating one feature partially activates others. Networks can mitigate these issues through non-linear operations (e.g. ReLU, softmax) that help disambiguate superposed features (Gurnee et al., 2023).

Definition 2 (Superposition Hypothesis) *A network layer represents features in superposition if:*

1. *The number of latent features exceeds layer dimensionality: $M > d_l$*
2. *There exists non-orthogonal feature directions: $\exists i, j$ with $i \neq j$ such that $\mathbf{v}_i \cdot \mathbf{v}_j \neq 0$*
3. *Latent feature activations are sparse: $\mathbb{E}_{\mathbf{x}}[\|\mathbf{a}(\mathbf{x})\|_0] \ll M$, where $\mathbf{a}(\mathbf{x}) = [a_1(\mathbf{x}), \dots, a_M(\mathbf{x})]^T$*

The network trades representational capacity against feature interference by packing more features than dimensions.

Adversarial attacks. Next we define the general notion of adversarial perturbations and attacks, following (Szegedy et al., 2014; Costa et al., 2024):

Definition 3 (Adversarial attack) *Let $\mathbb{B}_p(\mathbf{x}, \epsilon)$ denote the ϵ -ball around $\mathbf{x} \in \mathcal{X}$ in the ℓ_p -sense, e.g. $\mathbb{B}_\infty$ is the ℓ_∞ -ball. Any small δ that p -perturbs $\mathbf{x}$ to create an **adversarial input** $\mathbf{x}' \in \mathbb{B}_p(\mathbf{x}, \epsilon)$ such that $f(\mathbf{x}') \neq f(\mathbf{x})$ (untargeted) or $f(\mathbf{x}') = y_{\text{target}}$ (targeted) is known to be an **adversarial attack**. ℓ_∞ - or ℓ_2 -norm attacks are created via the projected gradient descent (PGD) (detailed in Sec. A).*

Definition 4 (Adversarial robustness) *A classifier $g_w(\cdot)$ with parameters w is said to be **robust** at $\mathbf{x}$ with radius $\epsilon > 0$, if $g_w(\mathbf{x}') = g_w(\mathbf{x}) \forall \mathbf{x}' \in \mathbb{B}(\mathbf{x}, \epsilon)$.*

3 SUPERPOSITION GEOMETRY CAN DETERMINE ADVERSARIAL ATTACKS

We begin by investigating if and how adversarial attacks exploit the interference rooted in the superposition of latent features. Specifically, we ask three questions:

1. *Do adversarial perturbations exploit the interference between superposed features?*
2. *Do correlations in the input shape the geometric arrangement of superposed latent features?*
3. *Can shared latent geometry explain why attacks transfer between independently trained models?*

Experimental setup. To investigate these questions, we design a synthetic task that exemplifies established LRH and superposition phenomena (Park et al., 2024; Elhage et al., 2022) in a precisely controllable setting, chosen so that the mechanisms we identify are general rather than tied to the idiosyncrasies of any one dataset. The task is designed to: (1) provide an intuitive classification setting for studying AExs; (2) represent class concepts as linear directions per the LRH; (3) induce controlled superposition between these latent features; and (4) retain a priori knowledge of how inputs correspond to the superposed features — enabling testable, transferable predictions about adversarial mechanisms.

Specifically, we partition input $\mathbf{x} \in \mathbb{R}^d$ into k equally-sized groups $\mathbf{x} = [\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)}]$, where $\mathbf{x}^{(j)}$ represents the p input features associated with class j . The task identifies which group has the largest sum: $y = \arg\max_{j \in \{1, \dots, k\}} \sum_{i=1}^p x_i^{(j)}$. Intuitively, each $x_i^{(j)}$ represents evidence for class j , with the p features capturing different class attributes. Following Elhage et al. (2022), we sample $x_i^{(j)} \sim \text{Uniform}(0, 1)$ with sparsity S (probability of setting to zero).

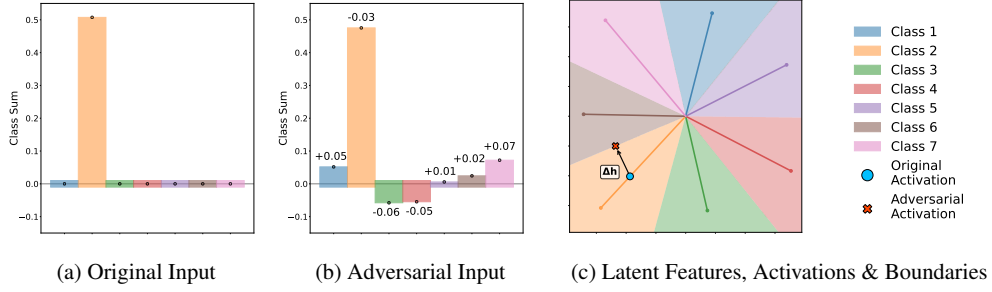


Figure 1: **An adversarial attack exploiting superposition geometry**, shown here for a representative configuration ($k = 7$, $m = 2$). (a) The original sample. (b) The adversarially perturbed sample, whose ground truth is unchanged yet which is misclassified. The sign and magnitude of each input perturbation are determined by the configuration of latent representations. (c) The original and adversarial sample in activation space. The arrows are the column vectors of $\mathbf{W}_e$, the latent representations of the input features.

A two-layer network compresses the input into m dimensions through a bottleneck encoder, $\mathbf{h} = \text{ReLU}(\mathbf{W}_e \mathbf{x} + \mathbf{b}_e) \in \mathbb{R}^m$ (where $m < k < d$), and reconstructs them via a decoder, $\mathbf{z} = \mathbf{W}_d \mathbf{h} + \mathbf{b}_d \in \mathbb{R}^k$. Our primary setup uses cross entropy (CE) loss without ReLUs or biases. This is chosen for its simplicity, with the softmax limiting interference by exponentially suppressing contributions from interfering latent features. To confirm that our conclusions are not artefacts of this particular configuration, we validate them under a second setup that uses mean squared error (MSE) loss with ReLUs and biases, and the findings hold across both. The representational challenge is to compress information from $d = k \times p$ dimensions to $m < k$ dimensions, and then recover sufficient information to identify the class with the largest sum.

Whereas non-synthetic tasks involve nuanced features that must be discovered and interpreted, this setup provides ground-truth class labels for the latent features. This lets us compute the true class after any perturbation by recalculating group sums, and grounds our definition of an AEx as satisfying the following two conditions:

1. The model’s prediction under perturbation changes (*e.g.* from class A to B)
2. The true class remains unchanged (*e.g.* still class A)

This distinguishes genuine AExs from perturbations that actually change the ground truth (*i.e.* change which group has the highest sum). Attacks are created using the two standard ℓ_∞ - and ℓ_2 -norm PGD threat models.

Evaluation metrics. We interpret the columns of the encoder weight matrix $\mathbf{W}_e$ as the latent representations $\mathbf{v}_i$ for each input feature x_i . Since input feature $x_i^{(j)}$ contributes exclusively to class j , these $\mathbf{v}_i$ (for $i \in \text{class } j$) align. This direct correspondence between input features and class-aligned latent directions lets us trace precisely how perturbations to the inputs of any given class, $\mathbf{x}^{(j)}$, relate to changes in latent space. For successful PGD-generated AExs $\mathbf{x}_{adv} = \mathbf{x}_{orig} + \delta$, we measure:

1. **Input perturbation profile (IPP):** The sign and magnitude of each δ_i , describing how different input features are perturbed in an attack.
2. **Latent attack alignment:** The similarity $\Delta \mathbf{h} \cdot \mathbf{v}_c$ between the *latent attack vector* $\Delta \mathbf{h} = \mathbf{h}_{adv} - \mathbf{h}_{orig}$ and class c ’s representation.
3. **Attack transferability:** Success rate of attacks generated on one model when applied to another.
4. **Robust accuracy:** The fraction of examples that remain correctly classified after perturbation δ with $\|\delta\|_p \leq \epsilon$ to a sample of inputs.

3.1 RESULTS

Do attacks exploit interference between latent features in superposition? Any attack must move a sample across a decision boundary to change its class, but what determines the required input perturbations? Two contrasting intuitions, each common in the broader literature, are available: the *feature intuition* suggests simply increasing the input features of the target class, whereas the *bug intuition* suggests adversarial perturbations modify inputs in essentially unpredictable ways, so that knowledge of feature representations offers no insight.

Fig. 1 shows a typical AEx in our setting: an input of Class 2 (orange) and its perturbed value (with IPP) that is misclassified as Class 6 (brown). The perturbations appear arbitrary – we clearly do not simply increase the target-class features, which contradicts the feature intuition. Yet they are not random either. Instead, they follow a precise pattern, with a systematic correspondence between δ and the configuration of latent features. This points to a third, more broadly applicable intuition: *attacks are mediated by latent geometry*.

Quantitatively, the IPP is strongly correlated with the latent attack alignment. The sign and magnitude of the perturbations $\delta^{(j)}$ track how closely their class’s latent representation $\mathbf{v}^{(j)}$ aligns with the latent attack vector $\Delta\mathbf{h}$. Positively aligned representations ($\mathbf{v}^{(j)} \cdot \Delta\mathbf{h} > 0$) have their input features proportionally amplified, while negatively aligned representations are attenuated. This is precisely what Fig. 1(b) illustrates – input perturbations set by the dot products in Fig. 1(c) so as to move the sample across the decision boundary.

To test whether PGD attacks specifically exploit superposition geometry, we compare PGD-generated attacks against theoretically optimal perturbations that we show leverage superposition (derived in Sec. 3.2). For each configuration (k, m) , spanning small to substantially larger settings, we train 5 models with different random seeds and generate 1000 PGD attacks per model, computing the cosine similarity between each successful perturbation and the corresponding optimum. We establish a random baseline by generating perturbations with matching ℓ_2 norm and measuring their similarity to the theoretical optimum. Tab. 1 shows near-perfect alignment between PGD and optimal attacks that persists across the full range of dimensionalities tested. We filter out samples with an ℓ_2 norm below ϵ to ensure meaningful perturbations. One-sample t-tests comparing observed similarities against the random baseline yield $p < 10^{-10}$ for every configuration, showing that PGD attacks systematically match the theoretical predictions rather than arising by chance. Since the optimal attacks leverage interference and PGD attacks align near-perfectly with these optima, we conclude that PGD exploits superposition interference.

Finding: Adversarial attacks systematically exploit interference between superposed features. Successful PGD attacks are predictable given the specific superposition geometry, rather than random.

Do input correlations determine latent feature geometry? We now ask whether correlations in the input data determine the geometric arrangements of latent features. By systematically varying the correlation structure of the training data, we test how these correlations constrain the degrees of freedom in the learned representations. We introduce three correlation patterns that span a spectrum of constraint strength. Under the *uncorrelated* condition, features are sampled independently, as in the previous subsection. For *paired* correlations, each feature in class i is coupled with a feature in class j , so that when one activates, so does its pair. For *global* correlations, each sample has a random phase that induces a cyclic activation pattern: classes adjacent in index have similar activation probabilities and thus frequently co-activate, while classes further apart rarely activate together. These three regimes are intended to bracket the range of correlation structures encountered in realistic data.

Fig. 2 shows how these correlations determine the arrangement of $\mathbf{v}^{(j)}$. We quantify geometric similarity by comparing the pairwise cosine-similarity matrices over all feature pairs in each model, then measuring the correlation between these matrices across random seeds; higher correlation indicates more similar geometric arrangements. For each correlation condition we tested 25 seed pairs. For the $m = 2, k = 7$ setup, geometric similarity was 0.23 ± 0.14 (uncorrelated), 0.47 ± 0.32

Table 1: Alignment between PGD-discovered attacks and theoretically optimal perturbations, reported across a range of superposition configurations. Results show mean $\pm$ std over 1000 attacks per condition. All p -values are below 10^{-10} .

k	m	Cosine Sim. (PGD vs Theory)	Cosine Sim. (Random Baseline)
6	2	0.97 ± 0.02	0.00 ± 0.02
30	10	0.96 ± 0.00	0.00 ± 0.01
90	30	0.92 ± 0.00	0.00 ± 0.00

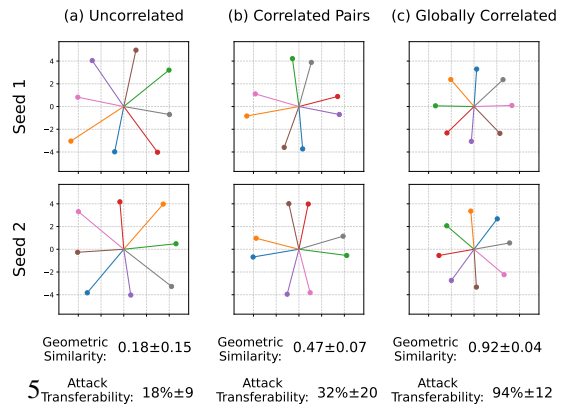


Figure 2: Greater input correlations yield more consistent geometries across initialisations, driving at-

(paired), and 0.88 ± 0.06 (global) (mean $\pm$ std). The results trace a clear monotonic relationship: uncorrelated data yields highly variable geometries across seeds, paired correlations partially constrain the arrangements, and global correlations force near-identical geometries. Statistical significance was assessed with pairwise two-sample t-tests between all three regimes under Bonferroni correction ($p < 10^{-3}$ for all comparisons, $\alpha = 0.05$ corrected for 3 comparisons). Results for a broad range of (k, m) configurations are reported in Sec. C.3, confirming the trend is not specific to any single setting.

The underlying mechanism is *interference avoidance*: frequently co-activating features are arranged to minimise mutual interference. Stronger correlations impose more constraints: uncorrelated data admits any arrangement that separates the classes, paired correlations fix the relative positions of coupled features, and global correlations determine a unique geometry up to rotation.

Finding: Input correlations constrain feature geometry. Stronger correlations reduce geometric degrees of freedom, forcing different initialisations to converge to similar arrangements.

Does shared geometry explain attack transferability? Having established that input correlations induce consistent latent geometries across model initialisations, we now test whether these shared interference patterns explain attack transferability more generally. If attacks are tailored to specific interference patterns, they should transfer successfully only between models with similar geometric arrangements.

Using the same seed pairs as above, we generate adversarial attacks on source models and evaluate their success when applied to target models within each correlation condition (25 transfer measurements per condition). All models achieve $> 95\%$ clean accuracy. Transfer rates correlate strongly with geometric similarity across the whole spectrum of conditions. For the $k = 7, m = 2$ configuration (Fig. 2), globally correlated data yields $94\% \pm 12\%$ attack transfer (mean $\pm$ std), uncorrelated data only $18\% \pm 9\%$, and paired correlations an intermediate $32\% \pm 20\%$.

The mechanism generalises naturally: each perturbation component amplifies or suppresses features, creating constructive interference that pushes representations across decision boundaries. When the same perturbation is applied to a model with different feature arrangements, features that previously interfered constructively instead cancel out or interfere destructively, so the attack fails.

Finding: Attack transferability is governed by shared interference patterns.

Does reducing superposition suppress these attacks? If superposition creates vulnerability through interference, then removing superposition should, in general, limit adversarial vulnerability. We test this prediction through three complementary experiments that each restrict interference. First, we set $m = k$, so the network learns to represent each class direction $\mathbf{v}^{(j)}$ orthogonally. In this configuration, any perturbation that changes the model’s prediction also changes the ground-truth class: moving a sample from class A to class B requires making the sum of class-B features exceed class A’s—genuinely transforming it into a class-B sample. We find zero successful adversarial examples across 1000 attempts at every ϵ value tested, showing that vulnerability scales with the degree of interference. Third, we test whether attacks concentrate on superposed features by constraining one class’s latent vector $\mathbf{v}^{(\perp)}$ to remain orthogonal to all others during training. When we then generate attacks between classes that remain in superposition (e.g. from class a to b where $\mathbf{v}_a \not\perp \mathbf{v}_b$), the inputs corresponding

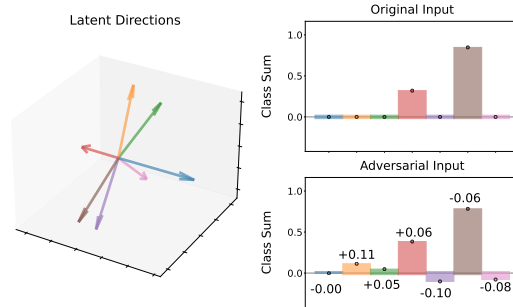


Figure 3: An adversarial attack (from class 5 to class 3) leaves untouched the input features of a class represented orthogonally (class 1), consistent with the general prediction that attacks concentrate on superposed features. Second, we isolate superposition’s effect from superposition pressure k/m (App. Tab. 5), showing that vulnerability scales with the degree of interference. Third, we test whether attacks concentrate on superposed features by constraining one class’s latent vector $\mathbf{v}^{(\perp)}$ to remain orthogonal to all others during training. When we then generate attacks between classes that remain in superposition (e.g. from class a to b where $\mathbf{v}_a \not\perp \mathbf{v}_b$), the inputs corresponding

to the orthogonal class stay unperturbed (Fig. 3). Together these controls point to a general conclusion rather than an isolated observation.

Finding: Adversarial attacks use their budget to exploit those features in superposition.

3.2 FORMAL ANALYSIS

We now formalise the mechanisms underlying our empirical observations, placing them on a footing that does not depend on the specifics of any one experiment. In our linear setting, we derive exact relationships between superposition interference and adversarial vulnerability, moving our results beyond observational correlation to a general characterisation. We adopt the notation from our experimental setup, where $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ induces superposition with columns $\mathbf{v}_i$, and the decoder corresponds to the encoder transposed (as empirically observed). The logit for class j is $z_j = \mathbf{v}_j^\top \mathbf{h}$.

Proposition 1. *The optimal input perturbations δ that maximise movement from class j to class k under constraint $\|\delta\|_2 = \epsilon$ satisfy $\delta \propto \mathbf{W}_e^\top \mathbf{n}$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ is the normal to the decision boundary between classes.*

Proof Sketch. To move from class j to k , we maximise the logit margin $z_k - z_j = \mathbf{n}^\top (\mathbf{h} + \Delta \mathbf{h})$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ and $\Delta \mathbf{h} = \mathbf{W}_e \delta$. This becomes $\max_{\|\delta\|_2 = \epsilon} \delta^\top \mathbf{W}_e^\top \mathbf{n}$. By Cauchy-Schwarz, this is maximised when $\delta \propto \mathbf{W}_e^\top \mathbf{n}$. Full proof in Sec. B. $\square$

Corollary 1 (Interference Drives Vulnerability) *The adversarial perturbation magnitude for feature i is $|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$, directly proportional to the differential interference between feature i and the class representations.*

This makes explicit how superposition creates adversarial vulnerability in general. Each input feature i is perturbed in proportion to its interference with the class representations. Under superposition, the non-orthogonality means that even semantically unrelated features bear on the class decision—and adversarial perturbations exploit exactly these dependencies to manipulate outputs.

Proposition 2. *Models with feature representations related by orthogonal transformation $\mathbf{Q}$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$) share identical optimal attack directions in input space.*

Proof Sketch. Under transformation $\mathbf{v}'_i = \mathbf{Q} \mathbf{v}_i$, optimal perturbations remain invariant: $\delta'_i \propto (\mathbf{v}'_i)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = \mathbf{v}_i^\top \mathbf{Q}^\top \mathbf{Q} (\mathbf{v}_k - \mathbf{v}_j) = \mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j) = \delta_i$. Full proof in Sec. B. $\square$

Together with our empirical findings, these propositions establish a **mechanistic pathway**. Specifically: (i) input correlations constrain feature arrangements in superposition, (ii) these geometric arrangements determine interference patterns, (iii) interference patterns dictate optimal perturbations via $\delta \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$ (Proposition 1), and (iv) shared geometric constraints yield similar interference patterns across models, enabling transferability (Proposition 2):

Correlations $\xrightarrow{\text{constrain}}$ **Feature Geometry** $\xrightarrow{\text{determines}}$ **Interference Patterns** $\xrightarrow{\text{enable}}$ **Transferability**

4 ATTACKS IN REALISTIC MODELS

We now extend our analysis from the controlled synthetic setting to more realistic models, taking a ViT (Dosovitskiy et al., 2020) trained on CIFAR-10 (Krizhevsky, 2009) as our primary case (and corroborating the trends on a ResNet-18 (He et al., 2016), Sec. D.2.1). By introducing a bottleneck layer directly before the final classification output, we again force the latent class features into superposition. We then examine: (i) how the degree of superposition affects attack transferability and robustness, (ii) whether a consistent arrangement of the superposed representations emerges across initialisations, and (iii) whether the attacks exploit feature interference—each a question we can now ask of a realistic architecture.

Setup. We train our ViT architecture (6 layers, patch size 4, residual stream $d = 512$) on CIFAR-10 across five random seeds. After training, we replace the original classification head with a bottleneck consisting of a linear encoder that compresses the pre-classification activations into an m -dimensional space, followed by a linear decoder that expands back to the $k = 10$ CIFAR-10 classes. This bottleneck, trained with CE loss and no ReLU (mirroring the setup in Sec. 3 so that results are directly

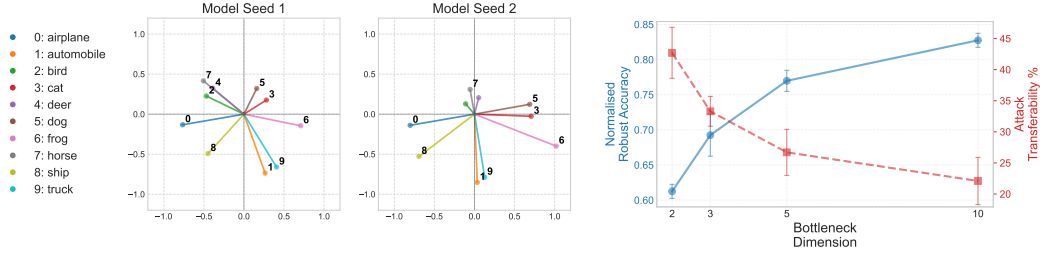


Figure 4: Left: the CIFAR-10 class-representation structure remains consistent across models trained from different seeds. Right: attack transferability and robust accuracy as the bottleneck dimension increases, exhibiting the same trends predicted by the toy analysis.

comparable), forces the k class representations into m -dimensional superposition. The weights of the base ViT are frozen while the bottleneck is trained. We sweep the bottleneck dimension $m \in \{2, 3, 5, 10\}$ to control the degree of superpositional compression. AExs are generated for each bottlenecked model using both ℓ_∞ - and ℓ_2 -norm PGD. We use a small step size and many iterations, terminating once an attack successfully changes the model’s prediction (full details in Sec. D.1). Robust accuracy is normalised by each model’s clean accuracy to account for performance variation across bottleneck capacities. Attack transferability is measured as defined in Sec. 3.

Results. The mean clean accuracy of the base models is 81%. As expected, the clean accuracy of the bottlenecked models decreases with compression: 54%, 67%, 76%, and 81% for $m = \{2, 3, 5, 10\}$.

We make three observations that mirror our toy-model results, providing direct evidence that the mechanism carries over to this more realistic setting rather than being confined to synthetic tasks. First, Fig. 4 (right) shows that as the bottleneck m decreases (*i.e.* superposition pressure increases): (i) normalised robust accuracy decreases, and (ii) attack transferability across different model initialisations increases. The decline in robust accuracy under greater superposition is consistent with the denser packing of class representations, which makes it easier for small input perturbations to cause widespread interference across the compressed features. The rise in transferability follows from our Sec. 3 finding that higher superposition reduces the degrees of freedom in the feature geometry. With a more constrained representational space, the network has fewer viable geometric arrangements for its class features, so different initialisations converge to more similar geometries and hence more shared interference patterns—yielding greater attack transferability. This same reasoning, established in the toy setting, thus predicts the behaviour of a full ViT. See Sec. D.1.1 for results across perturbation magnitudes.

Supporting this, Fig. 4 (left) shows that the geometric arrangement of the m -dimensional class representations within the bottleneck exhibits consistent ordering and clustering across models. This consistency, echoing the findings of Sec. 3, indicates that inherent correlations guide the formation of a superposed geometry in the realistic setting just as they do in the synthetic one. Notably, these groupings often reflect semantic similarities between classes (*e.g.* ‘cat’ and ‘dog’ representations clustering together), suggesting the mechanism engages meaningful structure rather than arbitrary artefacts. Finally, when AExs generated for a bottlenecked model are passed through the corresponding base ViT, they produce similar relative logit changes (*i.e.* $\text{logit}_{\text{base}}(\mathbf{x}_{\text{adv}}) - \text{logit}_{\text{base}}(\mathbf{x}_{\text{orig}})$). Both models therefore respond consistently to the same perturbations, indicating that these perturbations are not merely an artefact of the bottleneck; without the added compression, however, they do not on their own induce misclassification.

5 ADVERSARIAL VULNERABILITY VIA ALGORITHMIC BRITTLNESS

Prior sections argued that feature superposition can *suffice* for adversarial vulnerability. Here we ask whether it is *necessary* by studying a setting in which features are represented orthogonally, not in superposition, yet remain vulnerable to attack. This case supports the broader thesis of the paper: that analysing feature representations is a generally useful tool for robustness analysis, applicable well beyond superposition. We ask three questions: *how do such attacks work mechanistically, can this understanding be used to construct attacks directly, without gradients, and does naive robustness certification alter the underlying failure mode or merely raise the required attack budget?*

Setup. We train a linear encoder + MLP on one-hot integers (a, b) to predict $(a + b) \bmod P$, a task chosen because its learned solution is well understood and thus a clean probe of representation-level vulnerability. As detailed in Sec. E and Nanda et al. (2023), the model discovers a trigonometric algorithm to solve this task. As part of this learned algorithm¹ the encoder projects inputs onto non-basis-aligned linear representations encoding $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at key frequencies $\omega_k = 2\pi k/P$. Ultimately, the MLP composes $\cos((a + b)\omega_k)$ via sum-angle identities to reach 100% accuracy. Crucially, these features are represented *orthogonally* (i.e. not in superposition). We generate AExs by relaxing the one-hot constraint and perturbing only input a with PGD, moving inputs off the training manifold.

Attacks exploit the learned key frequencies. Fourier analysis of successful perturbations δ (App. Fig. 6) reveals that the dominant frequencies precisely match the learned algorithm’s key frequencies, ω_k . Consistent with the representation-level account, none (0%) of the attacks transfer to models with different ω_k , e.g. those trained from different seeds. The frequency-aligned perturbations cause misclassification by effectively altering the sinusoidal values read off from the encoder features. The perturbation δ transforms one-hot $\mathbf{x}_a$ into a dense vector $\mathbf{x}'_a$. The introduced contributions from the now non-zero inputs corrupt the sinusoidal values, thereby disrupting the MLP’s subsequent computations and leading to misclassification. As the model’s algorithm uses the constructive interference of each ω_k at the logits, disrupting a single ω_k is insufficient. Successful attacks must find δ that maximally alters multiple frequency components. We leave a detailed intuition of the mechanism to Sec. E.1

Analytic construction of attacks. Using this mechanism, we can construct gradient-free, *informed* perturbations built from composite sinusoidal waves of matching ω_k . On $0 + 0 \bmod P$, these attacks succeed with $\|\delta\|_2 = 0.22$ —close to PGD (0.17, $\approx 1.3\times$ the PGD budget) and far below naive alternatives (single-index 0.97, ℓ_∞ noise 2.04, uniform 6.20). This is empirical evidence that we are attacking the correct mechanism, and it illustrates a broadly applicable point: a representation-aware view of vulnerability can yield concrete attack recipes, not merely post-hoc explanations.

Do defences mitigate this algorithmic vulnerability? We apply certified training against ℓ_∞ perturbations (Müller et al., 2023). For fixed attack budgets, the *robust* accuracy improves from pre-robustified $\rightarrow$ post-robustified as follows: 0% $\rightarrow$ 85% at $\epsilon_\infty = 5e-3$, 0% $\rightarrow$ 96% at $2e-3$, and 76% $\rightarrow$ 100% at $1e-3$. However, when attacked with PGD, the robust model still fails via the same frequency-based brittleness, merely requiring larger perturbation norms. Certification therefore raises budgets without addressing the underlying fragility—a limitation we expect to generalise—motivating defences that explicitly target the identified semantics (here, frequency manipulation) rather than relying solely on broad norm bounds.

Relation to the paper’s thesis. This section makes two points that broaden the paper’s narrative. First, superposition is *sufficient but not necessary*: orthogonally represented features can still be vulnerable to attack, so the representation-level perspective encompasses more than superposition alone. Second, viewing AExs through the lens of *architecture–semantics interaction* is actionable across settings: it can guide both the design of attacks (via representation-informed perturbations) and the development of defences.

Although secondary here, we note a shared dependence on sparsity assumptions that hints at a unifying view: upstream violations of sparsity (e.g. noise or superposed features) can deliver effectively ‘pre-perturbed’ inputs to downstream orthogonal circuits, activating the same frequency-based failure.

6 RELATED WORK

Superposition & latent representations. A growing body of superposition research identifies feature-interference mechanisms that appear across models: Nanda (2023) distinguish *representational* and *computational* superposition, and Gurnee et al. (2023) differentiate *alternating* and *simultaneous* interference. Data correlations broadly shape these arrangements: pairs of correlated features become orthogonal (Elhage et al., 2022), drive superposition formation (Chan, 2024), and produce semantic

¹We refer to the specific computation executed by the network to solve the task as the *model’s learned algorithm*. Because this algorithm depends on extracting precise values from learned feature representations (frequencies), it remains *brittle* in the face of small perturbations to the bases. We term this *algorithmic brittleness*.

clustering (Prieto et al., 2025a). Beyond superposition, representation geometry is shaped by: spectral bias (Rahaman et al., 2019), neural collapse (Kothapalli, 2023), and optimisation objectives (Casper, 2023). Sparse autoencoders (SAEs) have been applied to vision models to disentangle features in superposition (Lim et al., 2024; Joseph et al., 2025).

Adversarial vulnerability. The broad literature is organised around an attack–defence dichotomy. Attacks (Szegedy et al., 2014) use imperceptible perturbations to maximise model error, including FGSM (Goodfellow et al., 2014) and PGD (Madry et al., 2018). Adversarial robustness concerns maintaining performance under such attacks. Defences include adversarial training (Madry et al., 2018) and certified training (Wong & Kolter, 2018; Palma et al., 2024), which provides guarantees against perturbations within specified input bounds.

The role of input & latent features on adversarial robustness. Across settings, vulnerability has been attributed to predictive but brittle ‘non-robust’ input features (Ilyas et al., 2019; Goh, 2019), or to learning shortcuts (Li et al., 2023), and perturbation characteristics are known to vary from one dataset to another (Maiya et al., 2021). Elhage et al. (2022) suggest superposition may be linked to AExs, a connection debated by Casper (2023). A general tension between representational efficiency and adversarial robustness has also been observed (Barsbey et al., 2025). Zhang et al. (2021) show that adversarial perturbations push latent representations across decision boundaries, and Ganeshan et al. (2019) find PGD targets final layers and propose feature-based attacks. Transferability is limited by representation discrepancies between models (Li et al., 2023; Wang et al., 2024); decorrelating features reduces transfer (Wiedeman & Wang, 2022), in line with our findings.

7 DISCUSSION & CONCLUDING REMARKS

We have shown that adversarial attacks can exploit interference patterns arising from the geometry of superposed features in NNs. Our experiments, spanning controlled toy models, ViTs, and additional architectures, establish that data properties—namely correlations and sparsity—induce distinct superposition geometries, which in turn create predictable adversarial vulnerabilities. These geometric arrangements and the resulting interference allow us to anticipate phenomena as varied as attack transferability and class-specific susceptibility. Taken together, this perspective frames adversarial vulnerability quite generally as a potential, inherent consequence of how networks efficiently encode large amounts of information through superposition, rather than as a quirk of any particular model or dataset.

Limitations & future work. Our insights are grounded primarily in simplified, controllable models, which we have used deliberately to isolate a mechanism we expect to recur more widely. While these analyses offer a mechanistic lens that we have shown transfers to larger models, the breadth of the adversarial-robustness literature suggests that the precise vulnerability mechanism may still vary with attack and model type. For instance, more constrained attack types (*e.g.* universal perturbations, jailbreaks, or patch attacks) may be unable to activate the specific feature combinations needed to exploit superposition interference, and characterising this boundary is an open question.

Promising future directions include quantifying the ‘adversarial cost’ of superposition, its connection to accuracy–robustness trade-offs, and how robust training reshapes superposition geometry. Because our controlled settings use class features in engineered superposition, whereas real networks involve interference among many unknown, unlabelled features across multiple layers, we plan to use SAEs (Bricken et al., 2023) to extend this analysis to large-scale networks; preliminary results in Sec. F indicate this is a promising direction toward a fully general account.

ETHICS STATEMENT

Owing to its investigative nature and broad, foundational focus, our mechanistic interpretability framework is not anticipated to have any specific negative impact or immediate societal consequences. Our experiments use only publicly available datasets (CIFAR-10) and synthetic toy models, involving no human subjects or personally identifiable information. The gradient-based attack methods used here are already well established in the literature; our work introduces no new attack capabilities but instead offers theoretical understanding of existing phenomena. We have no conflicts of interest to declare. By elucidating the general relationship between superposition and adversarial vulnerability, we hope to inform research directions that balance model capability with robustness, ultimately contributing to more trustworthy AI systems that can be safely deployed across a wide range of critical applications.

REPRODUCIBILITY STATEMENT

We include source code for all experiments in the supplementary materials. The toy model experiments (Sec. 3) come with complete architectural details, training procedures, evaluation metrics, and data-generation processes. For the ViT experiments (Sec. 4), architecture and training procedures are provided, with further details in Sec. D.1. The modular-arithmetic experiments (Sec. 5) detail the network architecture, data encoding, and learned trigonometric algorithm, with additional details in Sec. E.2. Experiments were conducted on an AMD Ryzen Threadripper 3970X 32-Core with 256GB RAM and an RTX 3090. Additional ResNet-18 and SAE experiments, which probe how broadly the findings extend, are documented in Appendices Sec. D.2.1 and Sec. F, respectively.

LLM USAGE STATEMENT

This work used LLMs to assist with prior literature search, to automate LaTeX table generation from CSV files (with all numerical values verified), and to suggest alternative phrasings that improve the clarity of the manuscript. All substantive research contributions and conclusions remain the original work of the authors.

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APPENDIX

A GENERATION OF ADVERSARIAL EXAMPLES

We generate AExs with PGD, a widely used iterative attack method (Madry et al., 2018). For untargted attacks, it maximises the classifier’s loss $\mathcal{L}(f(\mathbf{x}'), y_{\text{true}})$; for targeted attacks, it minimises $\mathcal{L}(f(\mathbf{x}'), y_{\text{target}})$, via the update rule:

$$\mathbf{x}'^{(k+1)} = \Pi_{\mathcal{S}} \left(\mathbf{x}'^{(k)} \pm \alpha \mathbf{g}_k \right) \quad (1)$$

(+ for maximisation, – for minimisation). $\mathbf{x}'^{(0)}$ is an initial perturbed input, α is step size, $\mathbf{g}_k$ is the (normalised) gradient $\nabla_{\mathbf{x}} \mathcal{L}(f(\mathbf{x}'^{(k)}), y_{\text{class}})$ (y_{class} being y_{true} or y_{target}), and $\Pi_{\mathcal{S}}$ projects onto the ϵ -ball $\mathcal{S} = \{\mathbf{x}' \mid \|\mathbf{x}' - \mathbf{x}\|_p \leq \epsilon\}$, with input domain clipping. We use the ℓ_p norms:

1. ℓ_{∞} : Constraint $\|\delta\|_{\ell_{\infty}} = \max_i |\delta_i| \leq \epsilon$. $\mathbf{g}_k$ is $\text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(\cdot))$. $\Pi_{\mathcal{S}}$ clips each $\mathbf{x}'_i$ to $[\mathbf{x}_i - \epsilon, \mathbf{x}_i + \epsilon]$.
2. ℓ_2 : Constraint $\|\delta\|_{\ell_2} = \sqrt{\sum_i \delta_i^2} \leq \epsilon$. $\mathbf{g}_k$ is $\nabla_{\mathbf{x}} \mathcal{L}(\cdot) / \|\nabla_{\mathbf{x}} \mathcal{L}(\cdot)\|_{\ell_2}$. $\Pi_{\mathcal{S}}$ rescales $\delta = \mathbf{x}' - \mathbf{x}$ if $\|\delta\|_{\ell_2} > \epsilon$ via $\delta \leftarrow \epsilon \cdot \delta / \|\delta\|_{\ell_2}$.

B OMITTED PROOFS & THEORETICAL RESULTS

To characterise, in general terms, how adversarial attacks exploit feature representations, we prove that optimal perturbations weight each input dimension by how strongly its corresponding feature aligns with the path to the decision boundary. We analyse our linear models without ReLU activation functions, a simplification that keeps the argument transparent while capturing the essential mechanism.

Consider input $\mathbf{x} \in \mathbb{R}^d$ encoded via $\phi(\mathbf{x}) = \mathbf{W}_e \mathbf{x}$ to latent representation $\mathbf{h} \in \mathbb{R}^m$, where the columns of $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ are an overcomplete dictionary $\{\mathbf{v}_i\}_{i=1}^d$ ($m < d$). For our argmax task, we empirically observe that the trained encoder-decoder pair has the decoder as the transpose of the encoder: $\mathbf{W}_d = \mathbf{W}_e^{\top}$. This means the logit for class j is computed as $z_j = \mathbf{v}_j^{\top} \mathbf{h}$, where $\mathbf{v}_j$ is the j -th column of $\mathbf{W}_e$. The predicted class is $\hat{y} = \arg \max_j z_j$. Input perturbations map to latent perturbations as $\Delta \mathbf{h} = \mathbf{W}_e \delta$, where $\delta = (\delta_1, \dots, \delta_d)^{\top}$ are the perturbation coefficients.

The decision boundary between classes j and k is the set of points where their logits are equal: $\mathcal{B}_{jk} = \{\mathbf{h} : z_j = z_k\}$, or equivalently $\{\mathbf{h} : (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{h} = 0\}$. The vector $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ acts as the normal to this pairwise decision boundary, pointing in the direction that maximally increases the margin $z_k - z_j$. We briefly recall the propositions before providing corresponding proofs.

Proposition 1 (Optimal targeted attack). The optimal input perturbations δ that maximise movement from class j toward class k under constraint $\|\delta\|_2 = \epsilon$ satisfy:

$$\delta \propto \mathbf{W}_e^{\top} \mathbf{n}$$

Proof. To move a sample from being classified as j to being classified as k , we need to maximise the logit margin $z_k - z_j$. Under perturbation δ , the new margin becomes:

$$z'_k - z'_j = \mathbf{v}_k^{\top} (\mathbf{h} + \Delta \mathbf{h}) - \mathbf{v}_j^{\top} (\mathbf{h} + \Delta \mathbf{h}) \quad (2)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^{\top} (\mathbf{h} + \mathbf{W}_e \delta) \quad (3)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{h} + (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{W}_e \delta \quad (4)$$

The change in margin is $(\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{W}_e \delta = \delta^{\top} \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)$. We seek:

$$\max_{\delta} \delta^{\top} \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j) \quad \text{subject to} \quad \|\delta\|_2 = \epsilon$$

Let $\mathbf{g} = \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)$. By the Cauchy-Schwarz inequality:

$$|\delta^{\top} \mathbf{g}| \leq \|\delta\|_2 \|\mathbf{g}\|_2 = \epsilon \|\mathbf{g}\|_2$$

Equality is achieved when δ and $\mathbf{g}$ are parallel. Given the constraint $\|\delta\|_2 = \epsilon$:

$$\delta = \frac{\epsilon}{\|\mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)\|_2} \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$$

□

Corollary (Interference drives vulnerability). For a targeted attack from class j to class k , the adversarial perturbation magnitude for input feature i is:

$$|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$$

where $\mathbf{v}_i$ is the i -th column of $\mathbf{W}_e$. This quantity represents the differential interference between feature i and the class representations.

This makes explicit the general mechanism by which superposition creates adversarial vulnerability. Each input feature i is perturbed proportionally to $\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)$ —the differential interference between feature i and the class representations. Under superposition, the non-orthogonality means that even semantically unrelated features have non-zero inner products with $(\mathbf{v}_k - \mathbf{v}_j)$, creating exploitable interference patterns. The multi-class setting only amplifies this vulnerability, since with k classes there are many possible pairwise boundaries, each with a distinct interference pattern. Adversarial perturbations leverage these cross-feature dependencies, manipulating features that bear on the decision through their interference with the class representations. This accounts for the vulnerability we observe empirically across settings: attacks succeed not by directly increasing target-class features, but by exploiting the web of interference created by superposition.

Prop. 1 characterises the optimal perturbation direction for moving from class j to k , giving the gradient that maximises the margin $z_k - z_j$. When intervening classes exist (where $z_j < z_i < z_k$ for some class i), following this gradient might cause the model to predict i before reaching the target k . Iterative methods such as PGD handle this by recomputing gradients at each step, so the global attack path emerges from repeated local decisions rather than a single optimisation. Moreover, although our analysis assumes $\mathbf{W}_d = \mathbf{W}_e^\top$ based on our empirical observations, the framework readily extends to arbitrary decoders. In the general case, with decoder $\mathbf{W}_d \in \mathbb{R}^{k \times m}$ having rows $\mathbf{w}_j^\top$, the optimal perturbation becomes $\delta \propto \mathbf{W}_e^\top (\mathbf{w}_k - \mathbf{w}_j)$, capturing the interference between encoder directions and decoder weights—so the characterisation is not tied to the tied-weights assumption.

Proposition 2 (Attack transferability). Consider encoders ϕ and ψ with basis matrices $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ and $\mathbf{W}'_e \in \mathbb{R}^{m \times d}$ whose columns are related by orthogonal transformation $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$). If both models use their encoder transpose as decoder (i.e., $\mathbf{W}_d = \mathbf{W}_e^\top$ and $\mathbf{W}'_d = (\mathbf{W}'_e)^\top$), then both models have identical optimal input perturbation vectors for any targeted attack from class j to class k .

Proof. For model ϕ , the optimal perturbation from Prop. 1 is:

$$\delta^\phi \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$$

For model ψ with transformed columns $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$:

$$\delta^\psi \propto (\mathbf{W}'_e)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = (\mathbf{W}'_e)^\top (\mathbf{Q}\mathbf{v}_k - \mathbf{Q}\mathbf{v}_j)$$

Since $\mathbf{W}'_e = \mathbf{Q}\mathbf{W}_e$ (all columns are transformed), we have:

$$\delta^\psi \propto (\mathbf{Q}\mathbf{W}_e)^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{5}$$

$$= \mathbf{W}_e^\top \mathbf{Q}^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{6}$$

$$= \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j) \tag{7}$$

where the last equality uses $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$. Thus δ^ϕ and δ^ψ have identical proportionality. Under the same norm constraint $\|\delta\|_2 = \epsilon$, we have $\delta^\phi = \delta^\psi$. □

Prop. 2 explains, in general, why attacks transfer between models with related training regimes. When models learn feature representations that differ only by an orthogonal transformation—essentially

the same geometric structure in a different orientation—they share identical vulnerability patterns in input space. The orthogonal transformation preserves all inner products between features and hence the interference patterns that attacks exploit. This accounts for our empirical observation, across both synthetic and realistic models, that networks trained on data with the same correlations exhibit high attack transferability: they discover similar feature geometries up to rotation, and therefore share adversarial vulnerabilities.

C TOY MODEL EXPERIMENTS

This section provides supplementary details, extended results, and further discussion for the toy-model experiments of the main paper. We report model accuracies across a much wider range of parameters than the main text, giving insight into how model capacity and data characteristics such as sparsity shape the learning process and the conditions under which feature superposition appears. We then present additional visual examples corresponding to Fig. 1, illustrating the mechanics of adversarial attacks under a variety of conditions and confirming that the behaviour is not specific to the settings highlighted in the main text.

C.1 HYPOTHESES TESTING FRAMEWORK

We now state explicitly the hypotheses corresponding to the three research questions posed in Sec. 3, framed so that each admits a general test rather than a setting-specific one.

Research Q1: Do adversarial perturbations exploit superposition geometry?

- H_0 : Adversarial perturbations are random with respect to feature geometry.
- H_1 : Adversarial perturbations systematically exploit geometric relationships between superposed representations.

Research Q2: Do data correlations determine superposition geometry?

- H_0 : Input correlations have no systematic effect on learned geometries.
- H_1 : Input correlations determine geometric arrangements across model initializations.

Research Q3: Does shared geometry explain attack transferability?

- H_0 : Attack transferability is independent of geometric similarity.
- H_1 : Transferability increases with shared latent structure.

We test these hypotheses through a set of controlled experiments designed to isolate the mechanism of interest:

- $H_1(1)$: We measure the input perturbation profile alignment with a class’s latent representation and the latent attack vector.
- $H_1(2)$: We systematically vary input correlations and measure resulting geometries.
- $H_1(3)$: We quantify transferability rates across models with varying geometric similarity.

C.2 ACCURACY OF TOY MODEL FOR A RANGE OF PARAMETERS

The toy-model experiments in the main paper predominantly used low-dimensional settings for conceptual clarity. To demonstrate that the model’s behaviour holds far more broadly, this subsection reports the classification accuracies achieved by the CE toy model across varying hidden-layer size (h), number of classes (k), number of features, and level of sparsity (S), giving insight into when the models learn to represent features in superposition. The sparsity level is the probability that any individual input feature $x_i^{(j)}$ is set to zero, so larger S means greater input sparsity. We report results as a function of feature density $1 - S$, the probability that a feature is non-zero. Tab. 2 and Tab. 3 together map out the model’s performance limits and its ability to learn latent representations in superposition over a wide parameter range.

Table 2: Classification accuracy of the CE toy model with a fixed bottleneck dimension ($m = 2$) across a broad range of class counts (k), total input features (features = $k \times 3$), and input feature densities ($1 - S$). These results illustrate the general way in which input sparsity governs performance degradation as more classes must be superposed within a constrained latent space.

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
3	9	2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
4	12	2	0.67	0.71	0.82	0.92	0.98	0.99	1.00	1.00
5	15	2	0.53	0.50	0.65	0.77	0.89	0.95	0.98	0.99
6	18	2	0.40	0.43	0.51	0.66	0.82	0.93	0.97	0.99
7	21	2	0.34	0.34	0.40	0.53	0.73	0.87	0.95	0.98
8	24	2	0.30	0.30	0.33	0.40	0.63	0.82	0.93	0.97
9	27	2	0.24	0.26	0.30	0.35	0.57	0.75	0.89	0.96
10	30	2	0.22	0.24	0.26	0.31	0.50	0.72	0.87	0.95
15	45	2	0.14	0.15	0.16	0.17	0.25	0.40	0.65	0.86
20	60	2	0.10	0.10	0.11	0.13	0.15	0.24	0.44	0.76
25	75	2	0.07	0.08	0.09	0.10	0.12	0.16	0.26	0.62
30	90	2	0.06	0.07	0.07	0.07	0.09	0.11	0.21	0.45

Table 3: Classification accuracy of the CE toy model across varying class counts (k), total input features, bottleneck dimensions (m), and input feature sparsity levels (S), extending the sweep to larger models. This table also reports different numbers of features per class (p).

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
30	30	30	0.23	0.24	0.38	0.62	0.83	0.94	0.99	1.00
30	90	90	0.27	0.27	0.33	0.41	0.51	0.67	0.85	0.94
40	40	30	0.67	0.54	0.73	0.77	0.88	0.96	0.99	0.99
40	120	30	0.71	0.64	0.65	0.72	0.73	0.79	0.89	0.96
60	60	10	0.05	0.07	0.12	0.25	0.47	0.73	0.90	0.97
60	180	10	0.08	0.10	0.13	0.17	0.22	0.32	0.52	0.75
80	80	30	0.15	0.17	0.23	0.41	0.63	0.79	0.91	0.98
80	240	30	0.23	0.22	0.31	0.41	0.48	0.53	0.66	0.80
100	100	10	0.03	0.04	0.05	0.10	0.21	0.43	0.69	0.87
100	300	10	0.04	0.05	0.06	0.09	0.12	0.15	0.25	0.45

C.3 CORRELATIONS CONSTRAIN ARRANGEMENTS

Tab. 4 reports the results for the experiments addressing *Do Input Correlations Determine Latent Feature Geometry?* across a wider range of class counts (k) and hidden-dimension sizes (m), confirming that the trend observed in the main text holds across configurations.

Table 4: Geometric-similarity results across correlation types and a range of superposition configurations, showing the monotonic trend is consistent throughout.

Correlation Type	k	m	Geometric Similarity
Uncorrelated	6	2	0.18 ± 0.15
	30	10	0.17 ± 0.02
	90	30	0.17 ± 0.01
Paired	6	2	0.47 ± 0.07
	30	10	0.26 ± 0.07
	90	30	0.25 ± 0.01
Global	6	2	0.92 ± 0.04
	30	10	0.88 ± 0.01
	90	30	0.80 ± 0.01

C.4 HOW THE TOY MODEL MAPS TO REAL-WORLD MODELS

How does scaling networks affect adversarial vulnerability? While empirical evidence suggests that larger models tend to be somewhat more robust to adversarial attacks, this effect is weak. Under adversarial training, clearer scaling trends emerge, but the improvements remain largely specific to the attack type used during training rather than conferring general robustness (Howe et al., 2025).

Two competing phenomena are at play when scaling up models. On one hand, larger models have greater capacity to represent concepts; on the other, there appears to be a long tail of useful concepts for a larger model to capture in general tasks such as next-token prediction over internet text. As a result, despite increased capacity, superposition appears to remain prevalent even in frontier models (Lindsey et al., 2025). Supporting evidence comes from dictionary-learning methods: SAEs require ever-larger dictionaries for larger models (Gao et al., 2025), indicating that the number of features scales with model size. The fundamental tension driving superposition—that models must compress many features into limited dimensions—therefore does not disappear with scale, and this reasoning is not confined to any single model family.

Since both superposition and adversarial vulnerability persist in large-scale models, we expect our insights to remain relevant across model scales and architectures. A promising future direction is to understand how the geometry of superposition changes with scale, which could in turn help mitigate vulnerability more generally.

C.4.1 SEPARATING SUPERPOSITION EFFECTS FROM CAPACITY REDUCTION

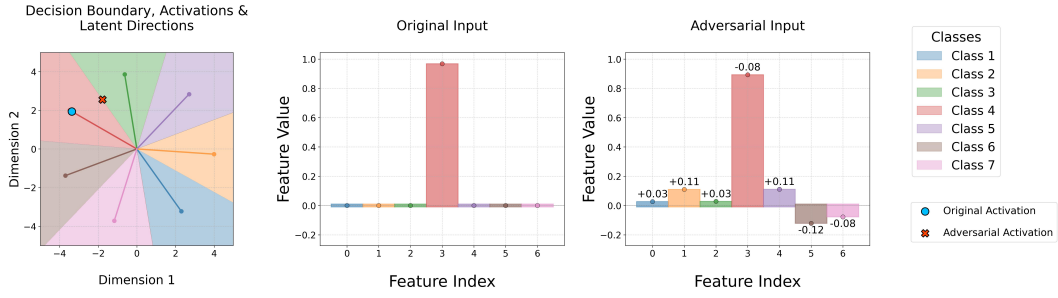
To disentangle superposition pressure from capacity reduction, we keep the bottleneck dimension fixed, which controls for the confound between the two. Fixing $m = 2$ and varying the number of classes k isolates superposition pressure. We report model accuracy, robust accuracy, and transferability in Tab. 5. These controls establish that increased superposition pressure (k/m), independent of model capacity, is what drives the adversarial vulnerabilities we observe—reinforcing that superposition, rather than capacity per se, is the general cause.

C.4.2 ADDITIONAL EXAMPLES OF AEXS IN TOY MODEL

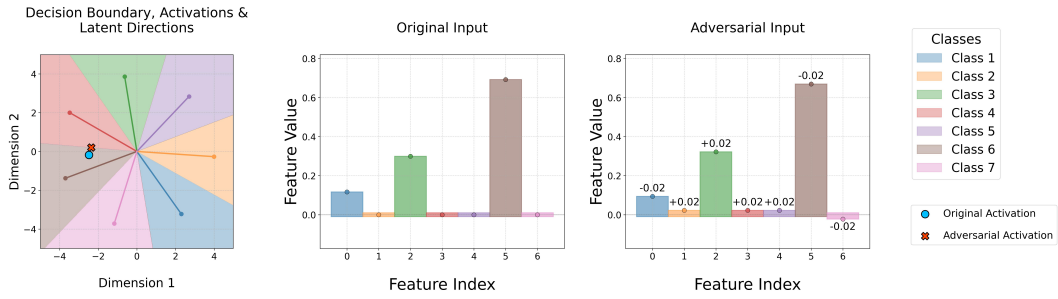
Fig. 1 illustrates how adversarial attacks exploit the interference between latent features in superposition. Here we provide further visual examples (Fig. 5) that reinforce this intuition and show it holds across attack norms and configurations. Specifically, we supplement the main text by showcasing:

Table 5: Performance metrics across correlation types, perturbation budgets (ε), and class counts (k), spanning a range of superposition-pressure conditions.

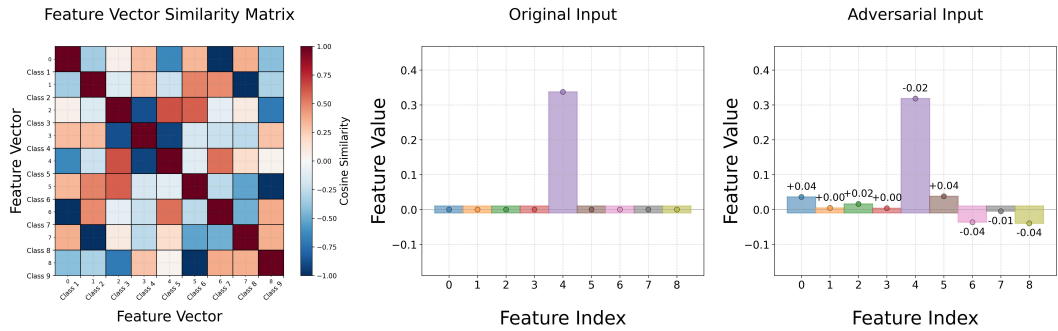
Correlation Type	Parameters		Performance (%)		
	ε	k	Accuracy	Robust Acc.	Attack Trans.
Uncorrelated	0.05	2	100.0	100.0	0.0
		4	100.0	94.9	21.2
		6	99.9	87.0	9.8
		8	99.8	77.7	17.5
	0.1	2	100.0	100.0	0.0
		4	100.0	88.1	9.8
		6	99.7	75.4	9.9
		8	99.6	56.3	23.5
	0.5	2	100.0	100.0	0.0
		4	100.0	99.4	0.0
		6	99.9	62.9	30.5
		8	99.5	24.9	39.0
Fully Correlated	0.05	2	100.0	100.0	0.0
		4	100.0	95.8	98.4
		6	97.3	70.6	98.4
		8	93.0	57.4	97.2
	0.1	2	100.0	100.0	0.0
		4	100.0	88.7	98.5
		6	97.7	45.6	99.7
		8	94.2	34.8	98.6
	0.5	2	100.0	100.0	0.0
		4	100.0	90.0	100.0
		6	97.2	34.6	100.0
		8	93.0	32.0	100.0



(a) An ℓ_2 -norm attack changing the classification of an input of class 4 to class 3. The left plot shows original and adversarial activations in latent space, along with representation directions. The right plots show original and perturbed input feature values.



(b) An ℓ_∞ -norm attack changing the classification of an input of class 6 to class 4. The left plot shows original and adversarial activations in latent space relative to class latent directions. The right plots show original and perturbed input feature values.



(c) An ℓ_2 -norm adversarial attack in a 7-class setup with an increased bottleneck dimension $m = 3$. The leftmost plot now shows the cosine similarity matrix between pairs of latent representations for each of the classes.

Figure 5: Visualisations of AExs in the toy model, supplementing Figure 1 of the main paper by illustrating the attack mechanism in both activation space and input space under a range of conditions, underscoring the generality of the pattern.

- Fig. 5a shows an additional instance of the setup in Sec. 3.1 ($m = 2, k = 7$) with an ℓ_2 -norm PGD attack, demonstrating the IPP and latent space manipulations that lead to misclassification.
- Fig. 5b shows a similar setup ($m = 2, k = 7$) using an ℓ_∞ -norm PGD attack on a less sparse input.
- Fig. 5c shows an example with increased bottleneck dimensionality ($m = 3, k = 7$) and accompanying ℓ_2 -norm PGD attack. The feature vector similarity matrix used to calculate geometric similarity is also shown.

D CIFAR-10 EXPERIMENTS

To test whether the principles observed in the toy models extend to more complex, realistic settings, Sec. 4 introduces experiments on CIFAR-10 (Krizhevsky, 2009) with an engineered bottleneck. This appendix section provides further details on the setup and presents extended results that reinforce the generality of the findings.

D.1 ARCHITECTURE & TRAINING INFORMATION

The base ViT (Dosovitskiy et al., 2020) comprised 6 transformer layers, an embedding dimension (d) of 512, and 8 attention heads per layer. Input images from CIFAR-10, sized 32×32 pixels, were split into 4×4 -pixel patches. The Multilayer Perceptron (MLP) within each transformer block had a hidden dimension of 512, and learned positional embeddings were used.

The bottleneck consisted of a linear encoder followed by a linear decoder. The encoder projected the pre-classification activations from the ViT backbone into an m -dimensional latent space ($m \in \{2, 3, 5, 10\}$), and the decoder mapped these m -dimensional representations back to the $k = 10$ CIFAR-10 classes.

The base ViT was trained on CIFAR-10 for 250 epochs with the Adam optimiser (default PyTorch parameters) at a learning rate of 0.001, a batch size of 512, a cosine-annealing learning-rate schedule, CE loss, and dropout. Training was repeated across five random seeds to account for variability. Once the base ViT was trained, its weights were frozen and the bottleneck was trained for 30 epochs at a learning rate of 0.001. For preprocessing, images undergo RandomCrop with 4-pixel padding, resizing to the target size (32×32 by default), and RandomHorizontalFlip for data augmentation.

D.1.1 NORMALISED ROBUST ACCURACY ACROSS PERTURBATION MAGNITUDES

Fig. 4 (right) shows how normalised robust accuracy varies with bottleneck dimension for a single value of ϵ . This subsection broadens that picture by reporting the normalised robust accuracy under PGD attacks of many different strengths. Results are given for both ℓ_2 -norm (Tab. 7) and ℓ_∞ -norm (Tab. 6) PGD attacks, providing a comprehensive view of how the degree of superposition interacts with attack strength to shape model robustness across the threat models considered.

AExs for these evaluations were generated with PGD using 100 iterations and a step size (α) of 0.01. Robust accuracy was evaluated on 500 samples per configuration, and we report the mean normalised robust accuracy and standard deviation across five random seeds.

Table 6: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for ViT models with different bottleneck dimensions (m) on CIFAR-10, subjected to ℓ_∞ -norm PGD attacks across a range of perturbation magnitudes (ϵ). Robust accuracy is normalised by the clean accuracy of each bottlenecked model. These results reinforce the findings of Sec. 4, with the same trend holding consistently across every ϵ .

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.001	96.0% $\pm$ 0.4%	97.4% $\pm$ 0.7%	97.9% $\pm$ 0.4%	98.1% $\pm$ 0.3%
0.01	61.7% $\pm$ 3.6%	69.6% $\pm$ 3.5%	77.0% $\pm$ 0.9%	81.8% $\pm$ 3.2%
0.05	4.9% $\pm$ 1.5%	6.7% $\pm$ 1.6%	9.3% $\pm$ 0.6%	10.5% $\pm$ 0.8%
0.1	0.1% $\pm$ 0.2%	0.2% $\pm$ 0.3%	0.2% $\pm$ 0.4%	0.2% $\pm$ 0.4%
0.5	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%

Table 7: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for ViT models with different bottleneck dimensions (m) on CIFAR-10, subjected to ℓ_2 -norm PGD attacks across a range of perturbation magnitudes (ϵ). Robust accuracy is normalised by the clean accuracy of each bottlenecked model. These results reinforce the findings in Section 4 of the main paper, with robustness decreasing consistently for smaller m (increased superposition) and larger ϵ .

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.1	90.4% $\pm$ 1.7%	91.8% $\pm$ 0.7%	94.6% $\pm$ 1.2%	95.0% $\pm$ 0.7%
0.5	58.5% $\pm$ 5.0%	60.8% $\pm$ 5.0%	69.2% $\pm$ 1.8%	72.6% $\pm$ 2.8%
1.0	41.7% $\pm$ 4.8%	44.0% $\pm$ 3.0%	50.4% $\pm$ 0.6%	54.9% $\pm$ 1.7%
2.0	34.0% $\pm$ 4.6%	36.2% $\pm$ 3.3%	41.5% $\pm$ 1.3%	47.4% $\pm$ 2.4%
5.0	30.2% $\pm$ 4.1%	32.9% $\pm$ 3.5%	37.3% $\pm$ 1.9%	42.7% $\pm$ 1.0%

D.1.2 ATTACK TRANSFERABILITY ACROSS PERTURBATION MAGNITUDES

We now report attack transferability across a broad range of perturbation magnitudes (ϵ) and bottleneck dimensions (m), so that the trend can be assessed under many conditions rather than a single one. Tab. 9 presents the ℓ_2 -norm attack transferability and Tab. 8 the ℓ_∞ -norm attack transferability.

D.2 CORRELATIONS IN CLASS FEATURES

In Sec. 3 it was correlations between inputs that drove the superposition arrangements. Here, we note that the arrangements are driven not by correlations between input classes but by correlations in the representations at this point in the network. At the classification layer this is likely analogous to the confusion matrix—*i.e.* how each class is misclassified relative to the others. Classes that cluster together are those the network finds inherently similar and misclassifies together. To test this more general reading, we repeat the experiment, fine-tuning the base ViT using timm/vit_base_patch16_384, which has been pre-trained on ImageNet-21k (14 million images, 21,843 classes) and ImageNet (1 million images, 1,000 classes). After fine-tuning on CIFAR-10 and applying the same bottleneck training procedure as Sec. 4, the resulting geometry shows random ordering across initialisations with approximately equal spacing between features (*i.e.* neural collapse Kothapalli (2023)). In this case performance is near 100%, so the confusion matrix is essentially the identity and the superposed structure is lost. In contrast, models trained solely on CIFAR-10 converge to similar geometries because they share the same learned difficulty structure: the same pairs of classes prove hard to distinguish, yielding consistent superposition patterns. This dependence on the difficulty structure, rather than on any surface property of the data, is what we expect to generalise.

D.2.1 RESNET-18

To probe architectural generality directly, we repeat the same experiments with a ResNet-18 (He et al., 2016) as the base model in place of a ViT. This convolutional architecture attains a slightly higher clean accuracy of 92% (versus 89% for the ViT). The robust accuracy falls slightly faster for ResNet-18 than for the ViT at matching ϵ values. Crucially, however, we observe the same qualitative behaviour—declining normalised robust accuracy and rising transferability as bottleneck pressure increases—indicating that the mechanism is not specific to transformer architectures.

E MODULAR ADDITION

Sec. 5 introduces a modular-addition task to study adversarial vulnerability in a scenario where features are orthogonal rather than superposed. This section provides a detailed experimental setup for the task, additional results, and further discussion of why we view this style of representation-level analysis as broadly useful beyond the superposition case.

Table 8: Attack transferability (%) for ℓ_∞ -norm PGD attacks on CIFAR-10 ViT models. Transferability is shown from a model trained with a specific ‘Source Seed’ (e.g., Seed 10) to three different target models, each trained with one of the seeds listed in the sub-header (e.g., ‘vs. Seeds 20/30/40’). The three slash-separated values in each cell give the transferability to these three target seeds, respectively. All models within a row share the same bottleneck dimension, m . The ‘Mean $\pm$ Std’ column averages transferability over all 12 source–target seed pairings for each (ϵ, m) configuration. Across this range of conditions, the results support the claim in Section 4 that higher superposition tends to yield more consistent latent geometries and thus higher transferability.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.001	2	77.8/66.7/44.4	25.0/62.5/25.0	72.7/72.7/72.7	58.3/75.0/58.3	59.3 $\pm$ 17.7
	3	57.1/28.6/28.6	14.3/28.6/28.6	63.6/72.7/72.7	42.9/42.9/42.9	43.6 $\pm$ 18.4
	5	70.0/50.0/50.0	57.1/71.4/57.1	50.0/25.0/25.0	16.7/50.0/33.3	46.3 $\pm$ 16.9
	10	55.6/55.6/66.7	12.5/37.5/25.0	16.7/33.3/16.7	28.6/71.4/28.6	37.3 $\pm$ 19.3
0.01	2	60.3/52.6/48.7	63.8/42.0/60.9	53.8/40.9/52.7	51.2/55.8/46.5	52.4 $\pm$ 6.9
	3	42.9/46.4/50.0	46.4/42.9/49.1	44.2/38.9/44.2	42.9/45.1/41.8	44.6 $\pm$ 3.0
	5	43.3/41.1/43.3	39.2/30.4/41.8	46.0/40.2/47.1	41.2/35.3/32.9	40.2 $\pm$ 4.8
	10	42.5/46.0/44.8	38.0/39.4/46.5	43.5/47.8/47.8	32.6/40.0/35.8	42.1 $\pm$ 4.7
0.05	2	35.0/36.4/37.8	38.8/36.0/46.3	37.4/31.3/41.7	42.0/42.0/40.7	38.8 $\pm$ 3.8
	3	29.9/29.9/37.1	30.4/30.1/35.6	30.2/30.2/35.1	30.1/31.9/34.4	32.1 $\pm$ 2.6
	5	26.8/25.3/29.2	25.2/21.5/26.4	28.0/28.0/26.5	26.0/26.6/21.9	25.9 $\pm$ 2.2
	10	27.1/24.9/30.4	21.6/23.0/28.4	25.3/28.1/30.6	21.0/24.9/20.1	25.4 $\pm$ 3.4
0.1	2	34.2/35.6/38.2	36.7/34.9/44.5	37.9/30.0/40.8	41.2/42.0/40.3	38.0 $\pm$ 3.8
	3	29.6/28.9/35.2	28.6/29.2/34.8	29.1/28.2/34.0	28.8/31.1/32.8	30.9 $\pm$ 2.5
	5	25.4/23.3/27.0	23.8/19.6/24.6	25.9/26.7/25.4	24.6/24.3/20.7	24.3 $\pm$ 2.1
	10	24.8/22.6/29.7	20.1/21.6/26.1	22.5/25.5/28.0	19.5/23.3/18.7	23.5 $\pm$ 3.3

E.1 LEARNED ALGORITHMIC REPRESENTATION IN MODULAR ADDITION

As detailed in Nanda et al. (2023) and Prieto et al. (2025b), the network discovers and implements a trigonometric algorithm to solve this task: the encoder projects inputs onto non-basis-aligned linear representations encoding $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at key frequencies $\omega_k = 2\pi k/P$. These features are represented orthogonally, *i.e.* not in superposition, and the sinusoidal values can be read off via linear probes. The subsequent MLP then applies sum-angle identities to compute $\cos((a+b)\omega_k)$ for various k , which constructively interfere at the logits to select the correct class. AExs are generated by relaxing the one-hot constraint and perturbing input $\mathbf{x}_j \in \mathbb{R}^P$ (encoding integer j) to $\mathbf{x}' = \mathbf{x} + \delta$ via ℓ_2 - and ℓ_∞ -norm PGD. For simplicity, only a is attacked while b is held fixed; the mechanism is symmetric in the two inputs.

As explained in Nanda et al. (2023), the network learns a structured trigonometric algorithm. W_e encodes each integer’s sine and cosine values for the key frequencies, and these features are linearly decodable sine/cosine values. Scaling the corresponding feature vector alters the encoded sinusoidal value itself, not its activation intensity. Integer embeddings W_e^j project onto a Fourier-like basis, giving the embedding space a circular/periodic structure. Unlike arrangements that arise from data correlations, this circularity stems from the sine/cosine encoding of the (uncorrelated) input integers—a distinct route to structured geometry that broadens the picture of how such geometry can form.

Our model learns to use approximately seven key frequencies (ω_k). Each embedding W_e^j for an input integer x_j predominantly encodes its fourteen corresponding trigonometric features: $\cos(\omega_i x_j)$ and $\sin(\omega_i x_j)$ for $i \in \{1, 2, \dots, 7\}$. The concatenated 200-dim. vector thus supplies these ≈ 28 key trigonometric features (fourteen for a and fourteen for b) to the MLP, which combines them via sum-angle identities (e.g., $\cos(A+B) = \cos A \cos B - \sin A \sin B$) to compute $(x_1 + x_2) \pmod{P}$.

Table 9: Attack transferability (%) for ℓ_2 -norm PGD attacks on CIFAR-10 ViT models. The table format, detailing source-model to target-model transferability, mirrors that of Tab. 8; see its caption for a full explanation. These ℓ_2 results, obtained under a different threat model, further support the claim in Sec. 4 that higher superposition leads to increased attack transferability.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.1	2	70.6/64.7/41.2	60.0/50.0/60.0	64.3/60.7/57.1	48.1/63.0/51.9	57.6 ± 8.0
	3	39.1/39.1/47.8	50.0/60.7/50.0	48.1/55.6/51.9	48.0/28.0/40.0	46.5 ± 8.4
	5	43.5/34.8/43.5	31.6/31.6/42.1	42.9/35.7/21.4	30.4/39.1/34.8	35.9 ± 6.4
	10	45.5/59.1/45.5	19.0/28.6/28.6	37.5/43.8/43.8	23.8/52.4/38.1	38.8 ± 11.4
0.5	2	59.6/52.1/48.9	60.8/35.4/60.8	54.1/39.4/51.4	43.8/49.5/46.7	50.2 ± 7.7
	3	38.7/40.6/48.1	36.1/36.8/41.7	37.4/36.5/43.5	34.4/34.4/36.7	38.7 ± 3.9
	5	38.4/32.0/37.6	29.4/25.7/32.1	39.1/33.9/38.3	31.8/29.9/32.7	33.4 ± 4.0
	10	38.3/37.4/37.4	26.5/28.4/31.4	32.4/35.3/35.3	23.4/28.2/25.0	31.6 ± 5.0
1.0	2	50.8/43.7/44.4	51.7/36.4/50.0	46.1/32.9/44.7	42.7/46.7/45.3	44.6 ± 5.3
	3	35.7/36.9/41.7	31.8/36.4/42.6	33.7/32.0/37.3	34.9/32.0/35.5	35.9 ± 3.3
	5	32.1/28.9/33.2	30.9/23.0/30.9	35.9/31.5/33.7	27.7/28.2/27.1	30.3 ± 3.3
	10	32.3/32.3/38.5	23.7/27.7/29.9	29.1/32.0/34.3	22.2/25.6/21.1	29.1 ± 5.0
2.0	2	47.0/41.6/44.3	48.2/36.2/48.2	43.9/33.5/42.1	43.4/43.4/42.8	42.9 ± 4.2
	3	31.9/37.7/40.3	31.1/35.6/40.6	32.6/31.1/36.8	35.4/33.3/34.3	35.1 ± 3.2
	5	29.9/25.4/32.1	26.3/22.1/26.7	32.3/31.8/29.5	27.9/27.9/23.6	28.0 ± 3.2
	10	30.8/30.4/35.7	22.5/24.9/28.6	24.8/29.7/32.7	20.4/26.1/19.0	27.1 ± 4.9
5.0	2	44.7/46.7/40.8	45.3/37.3/48.7	42.7/33.9/41.5	42.3/42.3/40.6	42.2 ± 3.8
	3	32.3/35.8/40.8	31.3/32.6/39.9	31.0/30.0/33.8	35.0/33.0/33.0	34.1 ± 3.2
	5	30.6/24.1/28.6	28.3/20.9/24.8	31.7/31.3/27.4	28.1/29.0/23.2	27.3 ± 3.3
	10	29.8/27.4/32.7	19.7/24.1/28.9	24.9/29.9/31.2	18.8/25.8/19.2	26.0 ± 4.6

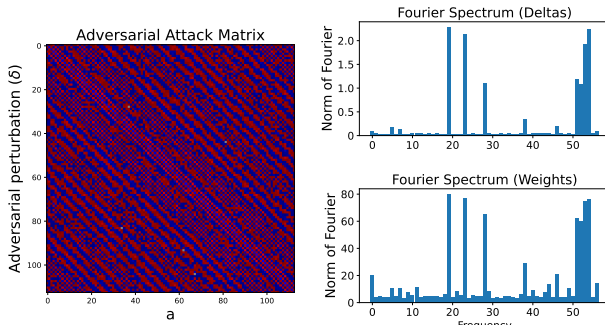


Figure 6: Left: each column is the mean attack δ added to a to successfully attack $a + b \bmod p$. Right: the norms of the Fourier components in δ and in $W_{MLP_{out}}$, showing the attack concentrates on precisely the learned key frequencies.

Table 10: Comparison of the minimum ℓ_2 -norm budget required to successfully attack $0 + 0 \bmod p$ across the different attack types considered.

Attack Method	Success ϵ
PGD	0.17
Informed Attack	0.22
Single a	0.97
ℓ_∞ Noise	2.04
Uniform δ	6.20

The algorithmic brittleness explored in Section 5 arises quite directly from adversarial perturbation of these key learned trigonometric features.

E.2 EXPERIMENT DETAILS FOR MODULAR ADDITION TASK

The network is trained for addition modulo 113. Inputs are integer pairs (a, b) , $a, b \in \{0, \dots, 112\}$. Each integer is mapped to a 100-dimensional vector via a shared embedding $\mathbf{W}_E \in \mathbb{R}^{113 \times 100}$, and the embeddings $\mathbf{W}_E^a$ and $\mathbf{W}_E^b$ are concatenated into a 200-dim. feature vector. This vector feeds a 3-hidden-layer MLP (200 ReLU neurons per layer), and the output layer produces 113 classification logits. The model is trained end-to-end with cross-entropy loss and AdamW (weight decay 4), reaching 100% test accuracy. AExs are generated with PGD under both ℓ_2 and ℓ_∞ norms across a range of ϵ . Attacks perturb the embedding $\mathbf{W}_E^a$ (with integer b fixed) by adding a perturbation δ so as to misclassify $(a + b) \pmod{113}$.

E.3 ALGORITHMIC BRITTLINESS

The frequency-aligned perturbations cause misclassification by effectively altering the sinusoidal values read off from the encoder features. To build intuition, we use the hypothesised neuron-aligned case: $\mathbf{W}_e$ would contain a row corresponding to the sine/cosine for each ω_k , whose element at column $\mathbf{w}_e^a$ holds the value $\sin(\omega_k a)$ or $\cos(\omega_k a)$. For input a , the one-hot $\mathbf{x}_a$ selects $\mathbf{w}_e^a$, giving $\mathbf{h}_a = \mathbf{w}_e^a$. The perturbation δ turns the one-hot $\mathbf{x}_a$ into a dense vector $\mathbf{x}'_a$, so that $\mathbf{h}'_a = \mathbf{W}_e \mathbf{x}'_a = (1 + \epsilon_a) \mathbf{h}_a + \sum_{k \neq a} \epsilon_k \mathbf{w}_e^k$. The contributions introduced by the $k \neq a$ sinusoidal terms corrupt the sinusoidal values, disrupting the MLP’s subsequent computations and leading to misclassification. Because the model’s algorithm relies on the constructive interference of each ω_k at the logits, disrupting a single ω_k is insufficient—the remaining components overwhelmingly support the correct answer. Effective attacks must therefore find δ that maximally alters several frequency components at once, a requirement we expect of any attack targeting such an algorithm.

E.4 ANALYTIC CONSTRUCTION OF ATTACKS AND BASELINE COMPARISONS

The main paper describes how understanding the learned trigonometric algorithm in the modular-addition task enables the analytic construction of adversarial attacks. Sec. 5 of the main paper benchmarks this ‘informed attack’ against PGD and several naive baselines. This section gives further detail on those baseline attack methods, which together span a spectrum from wholly uninformed to representation-informed. The baselines are:

- **PGD Attack:** Standard PGD used to find the adversary.
- **Informed Attack:** The analytically constructed perturbation, from combinations of cosines.
- **Single a :** This involved finding the minimal attack to a single input to cause misclassification.
- **ℓ_∞ Noise:** This baseline involved adding random noise δ_{noise} to the input a . Each component of δ_{noise} was sampled independently and uniformly from $[-\epsilon, \epsilon]$. The ϵ that successfully induced a misclassification is reported.
- **Uniform δ :** The perturbation vector δ is constrained to have all components equal, i.e., $\delta_j = c \cdot 1$, where c is a scalar. The magnitude of c is reported such that adding δ to the input causes a misclassification.

E.5 DETAILS ON CERTIFIED ROBUST TRAINING

To assess the impact of defences on the identified vulnerability, we performed certified robust training aimed at making the network provably robust against ℓ_∞ -norm perturbations. The certified training method was an abstraction-based technique (Wong & Kolter, 2018; Raghunathan et al., 2018; Palma et al., 2022), namely RSIP-IBP (Zhang et al., 2020; Henriksen & Lomuscio, 2023). RSIP-IBP combines Interval Bound Propagation (IBP) (Katz et al., 2017; Ehlers, 2017) and Reversed Symbolic Interval Propagation (RSIP) to compute reachable output sets for network layers, enabling formal certification of robustness within a given perturbation budget. The training loss incorporated both a standard component and an RSIP component. Certified training was run for 1000 epochs, with the first 200 serving as a warm-up phase, at a learning rate of 1×10^{-3} and weight decay 1×10^{-4} , using the same train/test splits as for standard training. The results, across a range of training perturbation budgets (ϵ_{train}) and evaluated against various attack budgets (ϵ_{attack}), are summarised in Tab. 11.

As discussed in Sec. 5, when these robustly trained models were subsequently attacked with PGD, the nature of the successful perturbations was unchanged: they continued to exploit the same key

Table 11: Verified robust accuracy (%) for modular-addition models under ℓ_∞ -norm perturbations. The table compares a standard (pre-robustification) model with models certifiably trained using RSIP-IBP at different training budgets (ϵ_{train}), evaluated against a range of attack budgets (ϵ_{attack}). These results support the discussion in Section 5.1 of the main paper: certified training improves robustness up to the specified ϵ_{train} , yet attacks continue to succeed by exploiting the same algorithmic sensitivities at higher ϵ_{attack} , a pattern we expect to hold generally.

Attack Epsilon (ϵ_{attack})	Train Epsilon (ϵ_{train})	Standard Acc. (Pre-Rob., %)	Robust Acc. (Pre-Rob., %)	Standard Acc. (Robust, %)	Robust Acc. (Robust, %)
5.0×10^{-4}	10^{-4}	100.0	100.0	100.0	100.0
	10^{-3}				100.0
1.0×10^{-3}	10^{-4}	100.0	76.0	100.0	93.8
	10^{-3}				100.0
2.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				95.7
5.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				85.3

frequency components inherent to the model’s learned algorithm. The main effect of certified training was to increase the perturbation magnitude required for misclassification at those budgets against which the model was successfully certified, rather than to change the attack vector’s alignment with the model’s algorithmic sensitivities. This suggests, more generally, that norm-based certification raises the cost of an attack without removing the underlying failure mode.

E.6 ON A POTENTIAL RECONCILIATION WITH SUPERPOSITION VULNERABILITY

Although the vulnerability explored here arises in the absence of feature superposition, it bears a striking phenomenological resemblance to those driven by superposition—a shared dependence on sparsity in the input representation. This motivates our *reconciliation conjecture*: that algorithmic vulnerabilities may persist in networks with orthogonal representations whenever sparsity assumptions are violated upstream. If earlier layers produce noisy or non-sparse features, potentially through adversarial interference, these deviations can propagate into otherwise well-isolated downstream circuits. Circuits that operate correctly on clean inputs may then fail, because the input they receive is effectively already perturbed. This offers a plausible, general route by which algorithm-disruption mechanisms, such as the frequency attacks studied here, could be triggered even without superposition. The broader question of how upstream superposition interacts with downstream algorithmic brittleness—and, more generally, a layer-by-layer understanding of the specific vulnerability mechanisms that AExs exploit—remains underexplored. We leave a detailed investigation, grounded in both empirical evidence and theoretical modelling, as a promising direction for future work toward a unified account.

F INSPECTING ADVERSARIAL PERTURBATIONS VIA SAE FEATURES

The analyses in the main paper focused on scenarios with either explicitly defined class features (the toy model) or class-level representations in a bottleneck (the ViT experiments). In large, practical networks, however, interference occurs between many unlabelled latent features across all layers. As noted in the future-work section of the main paper, SAEs offer a promising unsupervised means of extracting more of these linear features from NNs (Bricken et al., 2023). This appendix presents a preliminary investigation into using SAE-extracted features to characterise how a large-scale ViT responds internally to AExs versus clean inputs. These initial results are intended to gauge the feasibility of the approach as a path toward a general, feature-level understanding of adversarial phenomena in large models, rather than to present a conclusive set of new findings. We first outline the setup and validate the SAEs’ behaviour on adversarial inputs, then present exploratory layer-wise feature-difference analyses.

F.1 EXPERIMENT SETUP

We analyse SAE-extracted features from the vision encoder of the LAION/CLIP-ViT-B-32-DataComp.XL-s13B-b90K model, a 12-layer ViT with $\approx 70\%$ zero-shot accuracy on ImageNet-1k Deng et al. (2009)—a substantially larger and more general model than those studied elsewhere in the paper. We use the open-source Prisma SAEs, which map the 768-dimensional activation space to a dictionary of 49,152 features ($64\times$ expansion) (Joseph et al., 2025). We examine SAEs trained on either mean patch-token or CLS-token representations, across all layers, and analyse features activating above a high threshold (0.1), to capture the most strongly activating features, as well as a low threshold (0.001). For our SAE feature comparisons we generate 10,000 original/successfully-attacked image pairs (ℓ_∞ -norm, $\epsilon = 0.02$). This setup enables an initial exploration of how adversarial perturbations manifest at the level of SAE features in a large model.

F.2 SAE VALIDITY FOR ADVERSARIAL INPUTS

Because AExs are out-of-distribution inputs and our aim is to use standardly trained SAEs to understand them, we first perform a basic validation. This subsection examines the effect on model accuracy and SAE reconstruction when SAEs are inserted at inference time, for both clean and adversarial inputs. In effect we are testing whether the SAEs process features relevant to the classification task even under adversarial conditions, before attempting to interpret differences in feature activations.

Table 12: Impact on ImageNet-1k classification accuracy when inserting a pre-trained SAE (one layer at a time) during inference, for original, ℓ_∞ adversarial ($\epsilon = 0.02$), and random noisy inputs. ‘Accuracy’ refers to the ViT’s top-1 accuracy. ‘Original input + SAE’ means the clean ViT activations at a given layer are passed through the SAE and its reconstruction is fed to the next layer. The slight ‘de-attacking’ effect (improved accuracy for ‘Adv. input + SAE’ vs. ‘Adv. input’) suggests SAEs may filter some adversarial noise, but the model still largely misclassifies, indicating that the salient adversarial changes responsible for misclassification are preserved and processed.

Metric	Layer 2	Layer 5	Layer 8	Layer 11
Accuracy				
Original input	1.00	1.00	1.00	1.00
Original input + SAE	0.99	0.99	0.99	0.86
Adv. input	0.00	0.00	0.00	0.00
Adv. input + SAE	0.02	0.01	0.01	0.11
Noisy input	0.98	0.98	0.98	0.98
Noisy input + SAE	0.98	0.98	0.98	0.85
SAE Improvement				
Original input	-0.01	-0.01	-0.01	-0.14
Adv. input	0.02	0.01	0.01	0.11
Noisy input	-0.00	-0.00	-0.00	-0.13

F.3 LAYER-WISE SAE FEATURE DIFFERENCES

This subsection presents initial exploratory visualisations comparing SAE feature activations across layers of the ViT for original, ℓ_∞ adversarial, and random noisy inputs. These visualisations are meant to highlight promising avenues for future, more in-depth study of how adversarial attacks distinctively alter sparse feature representations in general. We examine:

- Activation count: Histograms comparing the number of SAE features activated above a set threshold in a layer for original, adversarial, and noisy images. This explores whether attacks characteristically alter feature sparsity.
- Feature overlap: The degree of overlap (number of common activated features) between (original vs. adversarial) and (original vs. noisy) conditions. This explores if attacks activate a distinctly different set of features.

- Jaccard Similarity: The Jaccard similarity between sets of activated features, offering a normalised measure of overlap.
- Mean activation: Histograms of mean activation values for features active above a threshold. This looks for changes in the intensity of (key) features.
- Distinct Features: The count of unique features activated across an image set for each condition.

F.4 CLS TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection presents a preliminary analysis of SAE features derived from CLS tokens, considering only features activated above a threshold of 0.1. Figures 7 through 10 visualise and compare, respectively, the histogram of activation counts, feature overlap, Jaccard similarity, and histogram of mean activations for original, adversarial, and random noisy inputs across the ViT layers.

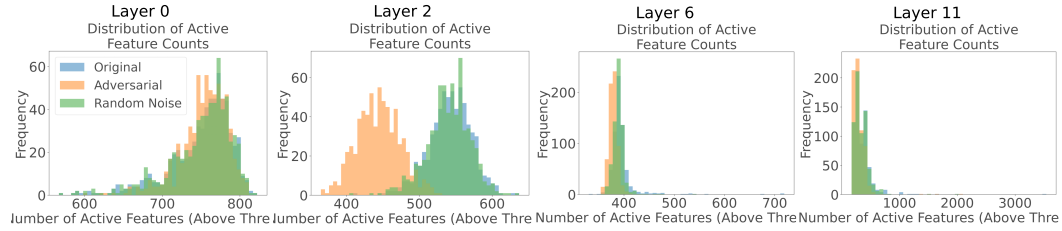


Figure 7: Histogram of activation counts for CLS token SAE features (threshold 0.1).

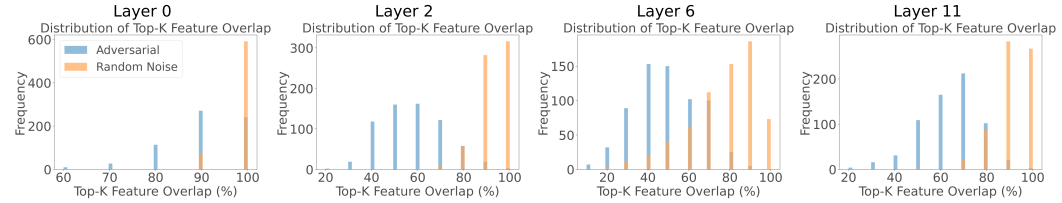


Figure 8: Feature overlap between original vs. attacked and original vs. noisy images for CLS token SAE features (threshold 0.1).

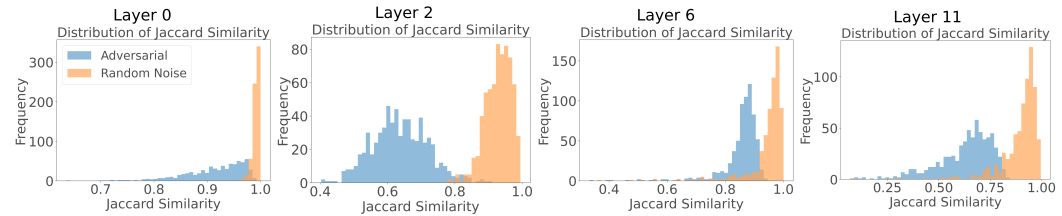


Figure 9: Comparison of Jaccard similarity for CLS token SAE features (threshold 0.1).

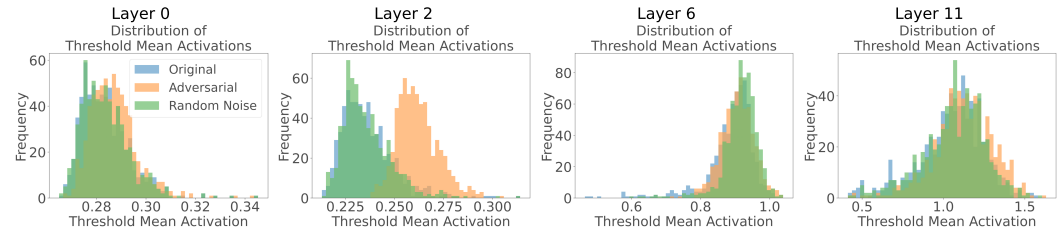


Figure 10: Histogram of mean thresholds for CLS token SAE features (threshold 0.1).

F.5 CLS TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

Here we examine SAE features from CLS tokens with a lower activation threshold of 0.001, so as to capture more than just the most intensely activated features. The subsequent figures illustrate the histogram of activation counts (Fig. 11), feature overlap (Fig. 12), Jaccard similarity (Fig. 13), and histogram of mean activations (Fig. 14) across layers for original, adversarial, and noisy inputs.

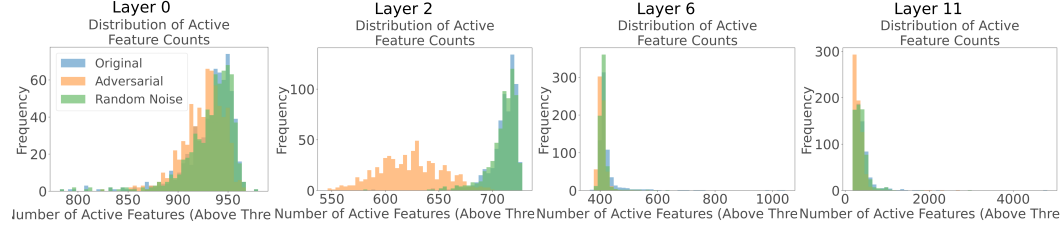


Figure 11: Histogram of activation counts for CLS token SAE features (threshold 0.001), with noise.

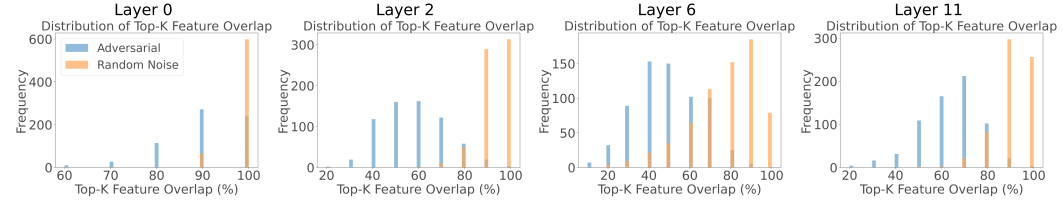


Figure 12: Feature overlap between original vs. attacked and original vs. noisy images for CLS token SAE features (threshold 0.001).

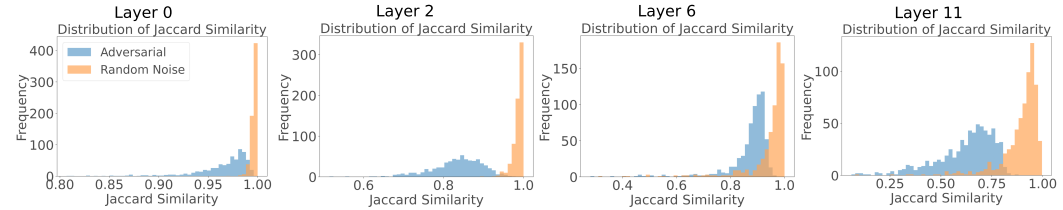


Figure 13: Comparison of Jaccard similarity for CLS token SAE features (threshold 0.001).

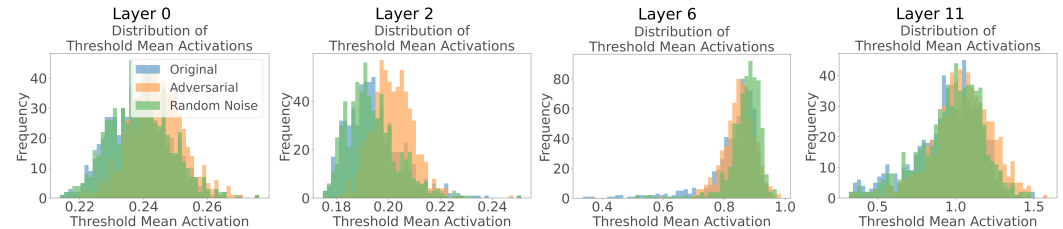


Figure 14: Histogram of mean activations for CLS token SAE features (threshold 0.001), with noise.

F.6 PATCH TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection shifts focus to SAE features derived from the mean of patch tokens, again using a high activation threshold of 0.1, to check that the observations are not specific to the CLS-token view. We present comparisons of activation-count histograms (Fig. 15), distinct-feature-count histograms

(Fig. 16), feature overlap (Fig. 17), Jaccard similarity (Fig. 18), and mean-activation histograms (Fig. 19) for original, adversarial, and noisy inputs.

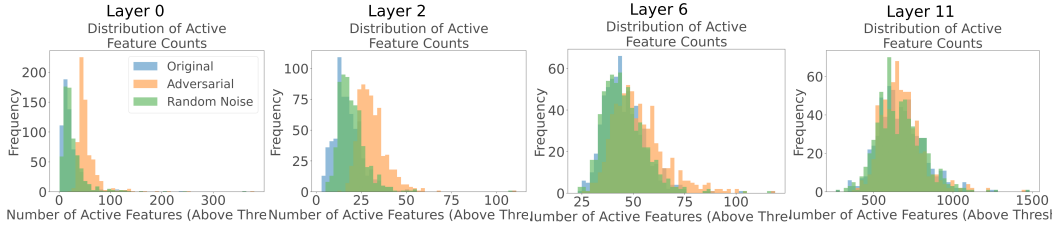


Figure 15: Histogram of activation counts for patch token SAE features (threshold 0.1), with noise.

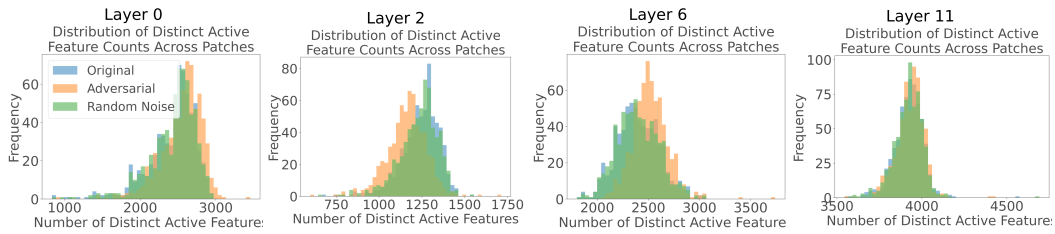


Figure 16: Histogram of distinct patch features for patch token SAE features (threshold 0.1), with noise.

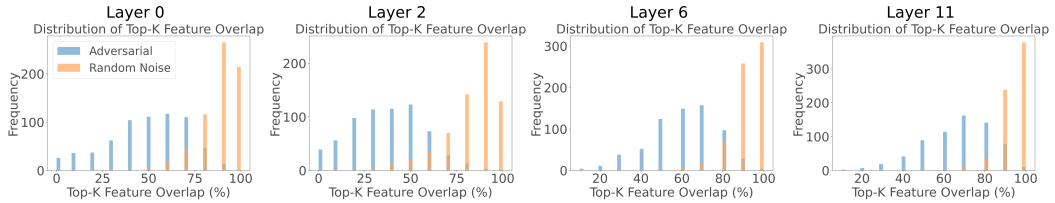


Figure 17: Feature overlap between original vs. attacked and original vs. noisy images for patch token SAE features (threshold 0.1).

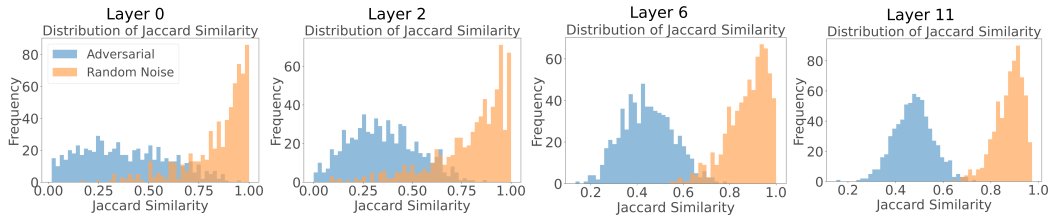


Figure 18: Comparison of Jaccard similarity for patch token SAE features (threshold 0.1).

F.7 PATCH TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

Finally, this subsection explores SAE features from the patch-token means using a low activation threshold of 0.001, completing the sweep over token types and thresholds. The figures show

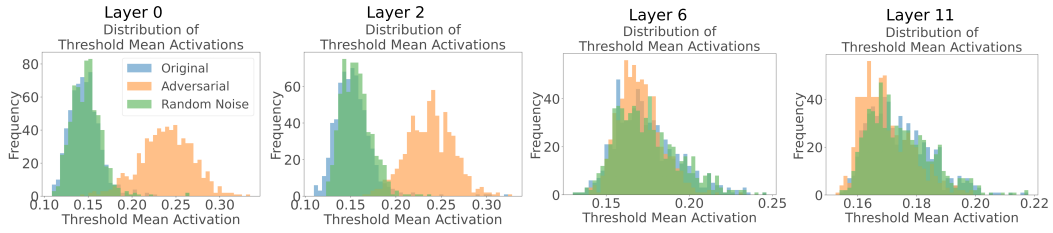


Figure 19: Histogram of mean thresholds for patch token SAE features (threshold 0.1), with noise.

histograms of activation counts (Fig. 20), distinct feature counts (Fig. 21), feature overlap (Fig. 22), Jaccard similarity (Fig. 23), and mean activations (Fig. 24) for original, adversarial, and noisy inputs.

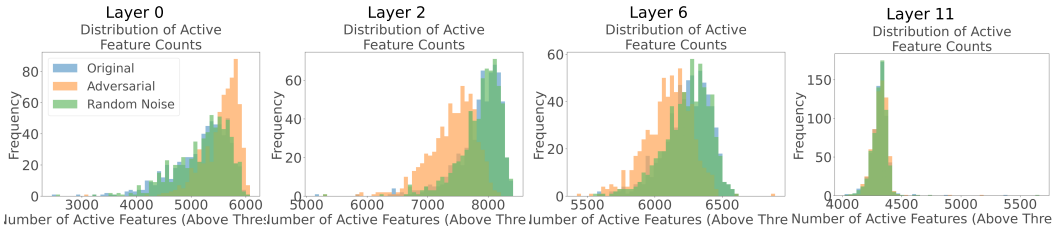


Figure 20: Histogram of activation counts for patch token SAE features (threshold 0.001), with noise.

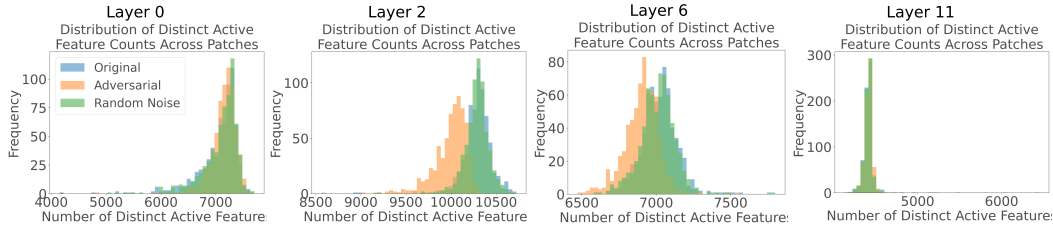


Figure 21: Histogram of distinct patch features for patch token SAE features (threshold 0.001), with noise.

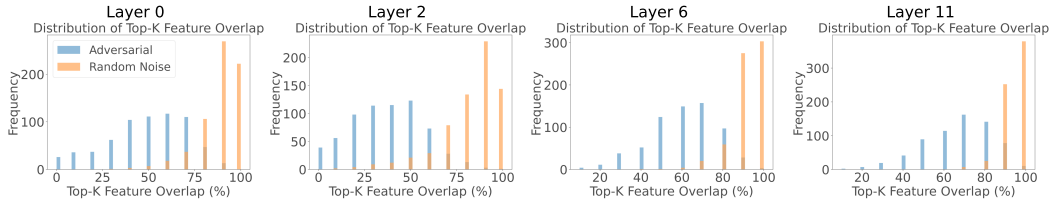


Figure 22: Feature overlap between original vs. attacked and original vs. noisy images for patch token SAE features (threshold 0.001).

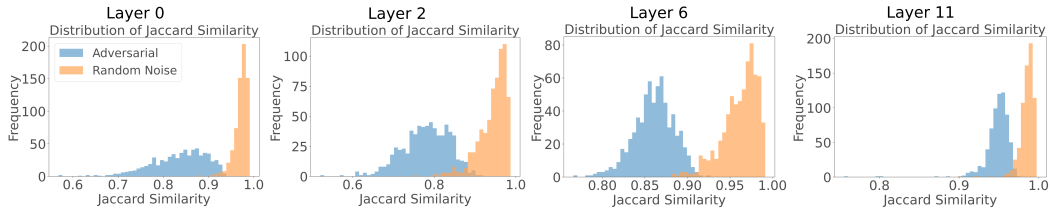


Figure 23: Comparison of Jaccard similarity for patch token SAE features (threshold 0.001).

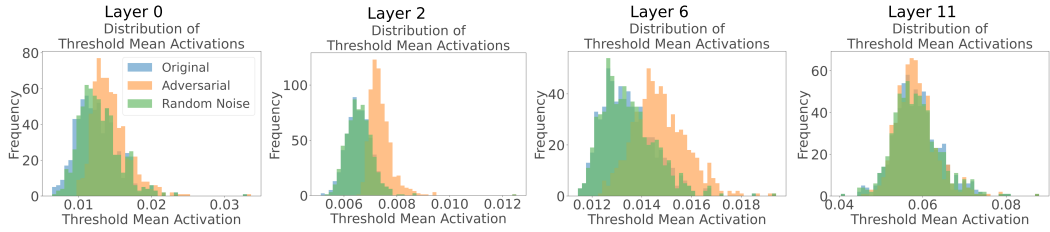


Figure 24: Histogram of mean activations for patch token SAE features (threshold 0.001), with noise.

G ACRONYMS

AEx adversarial example

CE cross entropy

LRH linear representation hypothesis

MLP Multilayer Perceptron

MSE mean squared error

NN neural network

PGD projected gradient descent

SAE sparse autoencoder

ViT vision transformer